One Anchor per Source: Hyperspherical Geometry and Causal Roles of the Latent Sources in an EEG/MEG Foundation-Model Tokenizer

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Abstract

How EEG/MEG foundation-model tokenizers organize information is unknown. We analyze the 16 latent sources of the BrainOmni tokenizer, whose quantizer receives unit vectors, in four public cohorts (316 participants; EEG and MEG; Parkinson's disease, dementia, depression). We describe each source in hyperspherical coordinates around its mean direction (anchor), prove when the anchor is the leading uncentered principal direction, and intervene with sphere-preserving edits and matched controls. Every source occupies a cap around a learned anchor shared across cohorts and modalities; it is the source's leading uncentered principal direction (cosine ≥ 0.997) yet carries only 1.3–4.9% of its centered variance. Moving a source away from its anchor damaged the reconstruction for all 64 source–cohort pairs, mainly through the decoder's attention keys, as did rotating it onto another source's anchor, whereas content edits acted through the values, mostly along a few

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leading tangent axes. The first quantization level encodes most of the source identity. Replicas trained from scratch reproduced the anchors, the reading of identity through the keys and this identity code, but read location through both keys and values; with a decoder source embedding (two runs), the anchors did not separate. In subject-level read-outs, the sources and their tokens carried diagnostic information wherever band power did; within cohorts the anchor coordinates did not significantly exceed random directions, but in transfer between two Parkinson's cohorts they did, in one or both directions depending on feature scaling. The anchors act as addresses, not carriers of content, which mean-direction and principal-component analyses miss.
Keywords: mechanistic interpretability, brain foundation models, EEG, MEG, residual vector quantization, hyperspherical geometry, causal intervention

1. Introduction

Foundation models for electroencephalography (EEG) and magnetoencephalography (MEG) are trained on thousands of hours of recordings and fine-tuned for clinical and cognitive decoding tasks (Jiang et al., 2024; Yang et al., 2023; Wang et al., 2025, 2024; Jiang et al., 2025; El Ouahidi et al., 2025). Several of them do not operate on raw signals but on discrete tokens produced by a learned tokenizer, typically a vector-quantized autoencoder (van den Oord et al., 2017; Jiang et al., 2024, 2025; Ma et al., 2026). The tokenizer fixes what the rest of the model can see: whatever it discards is unavailable downstream, and whatever structure it imposes is inherited by every model built on its tokens. Understanding how a tokenizer organizes its

internal representation is therefore a prerequisite for interpreting, compressing or clinically deploying such models.

BrainOmni (Xiao et al., 2025) is a foundation model for EEG and MEG whose tokenizer has an unusual and interpretable bottleneck. Instead of quantizing sensor signals directly, its encoder pools all sensors into 16 *latent sources* through cross-attention with learned queries, L2-normalizes each 256-dimensional source vector and quantizes it with a four-level residual vector quantizer (RVQ) (Juang and Gray, 1982; Zeghidour et al., 2022; Défossez et al., 2023). The decoder reconstructs every sensor from the quantized sources and receives no source index, so it can tell the sources apart only through their quantized vectors. Because the quantizer only ever sees unit vectors, the representation that matters is the direction of each source vector on the unit hypersphere. How these directions are distributed, which aspects of them the decoder relies on, and what they reveal about the recorded person are open questions.

Answering them requires care, because the standard tools of representation analysis have known failure modes in exactly this setting. Learned representations are often dominated by a common mean direction (Mu and Viswanath, 2018; Ethayarajh, 2019; Gao et al., 2019; Biś et al., 2021), which makes centered and uncentered principal component analysis (PCA) (Jolliffe and Cadima, 2016) answer different questions and makes mean directions look like directions of large variation when they are not. Projection (zero) ablation conflates the removal of a constant offset with the removal of information, which is why mean and resample ablation and optimized ablation have been proposed as alternatives (Chan et al., 2022; Heimersheim and Nanda, 2024;

Zhang and Nanda, 2024; Li and Janson, 2024). Linear probes measure decodability rather than use (Hewitt and Liang, 2019; Belinkov, 2022; Elazar et al., 2021), and probing results on small neuroimaging cohorts are unstable, biased by leave-one-subject-out designs and easily confounded by subject identity (Varoquaux, 2018; Airola et al., 2011; Saeb et al., 2017; Lin et al., 2026).

This paper analyzes the latent sources of BrainOmni's tokenizer with methods designed around these pitfalls. We describe the unit vectors that the quantizer receives in hyperspherical coordinates relative to each source's mean direction; intervene on them with edits that never leave the sphere and compare these with controls matched in geodesic displacement, dimension and *support* (the span of the 64 leading tangent axes, which contains almost all of the variation of a source's vectors around its anchor); split the effect of an edit into the two pathways through which the decoder's attention reads the sources, its keys and its values; trace edits through the resulting tokens; and evaluate diagnostic content at the level of subjects with permutation inference. The analysis covers four public cohorts with 316 participants in total, two of them with Parkinson's disease, one with Alzheimer's disease and frontotemporal dementia and one with adolescent major depression, recorded with 19- to 60-channel EEG or 269-channel MEG; and it covers the released checkpoint, the copy embedded in the BrainOmni foundation-model checkpoints (the same encoder and decoder with other codebooks), and four tokenizers trained from scratch, two of them with a learned source embedding in the decoder, which tests whether the structure we describe arises because the quantized vectors are the decoder's only access to source identity.

Contributions.

1. **Hyperspherical description and its relation to principal components.** We describe each latent source by its *anchor* (mean direction), the *anchor coordinate* $a = \cos\theta$ of each vector, with θ its *colatitude*, and its *tangent direction*, and show, from elementary identities and the residual form of the Davis–Kahan theorem, when the anchor is the leading eigenvector of the uncentered second moment. The anchor satisfies the eigenvalue equations of both second moments up to the same residual, which explains how it can be the leading uncentered principal direction of every source (cosine ≥ 0.997) while carrying only 1.3–4.9% of the centered variance and spreading over several centered principal components.
2. **Sphere-preserving interventions.** Edits that map the unit sphere onto itself, so that the quantizer's normalization cannot undo them, separate the location, content and identity of a source. They come with cross-fitted statistics, controls matched in displacement, dimension and support, and a split of the decoder's attention into its keys and its values.
3. **A two-part code.** Every source occupies a cap around a learned anchor that is nearly the same across cohorts, sites, montages and modalities. Where the vectors lie relative to the anchors is read mainly through the decoder's attention keys, the content lies along a few leading tangent axes and is read through the values, and the first token level carries most of the information about source identity.
4. **Replication and a test of the mechanism.** Two tokenizers trained

from scratch reproduce the anchors, the reading of identity through the decoder's keys, the content along leading tangent axes read through the values, and the identity code of the first token level, although their decoders read the location of a source through both pathways. Two variants whose decoder receives a learned embedding of each source do not separate the anchors, read location and content almost only through the values, and encode little identity in their first token level.

5. **Diagnostic content.** In subject-level read-outs with permutation inference across five contrasts, the source representation and its tokens carry diagnostic information wherever band power does. Within a cohort the 16 anchor coordinates are not a privileged summary of it, and between two Parkinson's cohorts they transfer better than random directions in one direction with the training cohort's feature scaling and in both with features standardized within each cohort.

Section 2 reviews related work, Section 3 describes the tokenizer, Section 4 defines the quantities, interventions and statistical procedures, Section 5 reports the results and Section 6 discusses their implications and limitations.

2. Related work

Brain foundation models and their tokenizers. Self-supervised models for EEG range from contrastive and masked-prediction transformers (Kostas et al., 2021; Yang et al., 2023; Cui et al., 2024; Wang et al., 2024, 2025; Tegon et al., 2025; Döner et al., 2025; Zhou et al., 2025; El Ouahidi et al., 2025) to models for intracranial recordings (Zhang et al., 2023; Wang et al., 2023a; Yuan et al., 2024). Tokenizer-based models learn a discrete vocabulary first:

LaBraM trains a vector-quantized neural spectrum predictor (Jiang et al., 2024), NeuroLM aligns such tokens with a language model (Jiang et al., 2025), CodeBrain decouples temporal and frequency codebooks (Ma et al., 2026), and single-channel motif tokenizers have been proposed (Pradeepkumar et al., 2026). BrainOmni (Xiao et al., 2025) unifies EEG and MEG through a sensor-agnostic latent-source bottleneck; its tokenizer is the object of this study. Evaluations of brain foundation models have so far focused on downstream accuracy, its dependence on participant diversity and its comparison with conventional decoders (Yang et al., 2026; Bomatter and Gouk, 2025; Kuruppu et al., 2025; Thölke et al., 2026).

Interpreting EEG foundation models. Recent studies apply sparse autoencoders (Bricken et al., 2023; Huben et al., 2024; Gao et al., 2025) to EEG foundation models (Lehn-Schiøler et al., 2026; Cao et al., 2026), probe and erase linear subspaces of their features (Tang et al., 2026), test their robustness and attributions (Širca et al., 2026), and show that subject identity can dominate frozen features on small clinical cohorts (Lin et al., 2026). These works analyze backbone features; we analyze the tokenizer bottleneck, whose geometry is fixed by an explicit normalization and quantization step and whose causal role can be measured exactly through its reconstruction.

Directions, ablations and probes. Interpretability work on language and vision models identifies concepts with directions (Park et al., 2024; Marks and Tegmark, 2024), tests them by activation patching and ablation (Vig et al., 2020; Meng et al., 2022; Wang et al., 2023b; Conmy et al., 2023; Geiger et al., 2024; Arditi et al., 2024), and decodes them with probes (Alain and Bengio, 2017; Hewitt and Liang, 2019; Belinkov, 2022). How a component is ablated

changes the conclusion: zero ablation moves activations off their distribution, which mean and resample ablation avoid (Chan et al., 2022; Heimersheim and Nanda, 2024; Zhang and Nanda, 2024), and optimal ablation formalizes the choice (Li and Janson, 2024). Probing results must be interpreted against control tasks and erasure (Hewitt and Liang, 2019; Elazar et al., 2021; Ravfogel et al., 2020; Belrose et al., 2023). Attention can be decomposed into a query–key pathway, which determines where information is read from, and a value–output pathway, which determines what is read (Elhage et al., 2021); we use this decomposition to locate the role of the anchors in the decoder. Our intervention family follows these principles and adds controls specific to normalized inputs: rotations matched in geodesic displacement and random subspaces matched in dimension and support.

Dominant directions and anisotropy. Word and contextual embeddings share a large common mean vector and a few dominant directions (Mu and Viswanath, 2018; Ethayarajh, 2019; Gao et al., 2019; Biś et al., 2021); transformers develop outlier dimensions and massive activations that act as biases (Kovaleva et al., 2021; Timkey and van Schijndel, 2021; Dettmers et al., 2022; Sun et al., 2024; Darcet et al., 2024). In all of these cases the uncentered and centered second moments differ by a rank-one mean term (Honeine, 2014), so the leading uncentered principal direction tracks the mean while centered PCA describes the variation around it. Eigenvector perturbation theory (Davis and Kahan, 1970; Yu et al., 2015) quantifies when the two coincide; we use it to make the relation between anchors and principal directions exact. Representations normalized onto the unit hypersphere are common in contrastive learning, where their distribution is described by alignment and uniformity (Wang and

Isola, 2020), and in vector quantizers with L2-normalized codes (Yu et al., 2022); directional statistics supplies the natural location and concentration parameters of such data (Mardia and Jupp, 2000).

Classical decompositions and clinical EEG. Independent and principal component analyses have long been used to decompose EEG (Makeig et al., 1995; Bell and Sejnowski, 1995; Hyvärinen and Oja, 2000), and band power is the standard feature for clinical EEG classification, including convolutional models (Schirrmeister et al., 2017; Lawhern et al., 2018). Diagnostic decoding from small cohorts requires subject-wise validation and calibrated chance levels (Saeb et al., 2017; Varoquaux et al., 2017; Varoquaux, 2018; Combrisson and Jerbi, 2015; Poldrack et al., 2020), avoidance of leakage (Kapoor and Narayanan, 2023; Brookshire et al., 2024), and awareness that leave-one-out estimates of the area under the receiver operating characteristic curve (AUC) are biased (Parker et al., 2007; Airola et al., 2011); permutation tests provide exact inference for cross-validated scores (Ojala and Garriga, 2010).

3. The BrainOmni tokenizer

Figure 1a summarizes the architecture.

Input. In the BrainOmni pipeline for downstream data, a recording with N_c sensors is notch filtered at the mains frequency, band-pass filtered (0.1–96 Hz), resampled to 256 Hz and cut into non-overlapping segments (10 s for EEG and 30 s for MEG; we used 10 s throughout, Section 4.1); in every segment, the across-sensor mean is subtracted at every sample and the data are divided by their global standard deviation, separately for each sensor type (Xiao et al.,

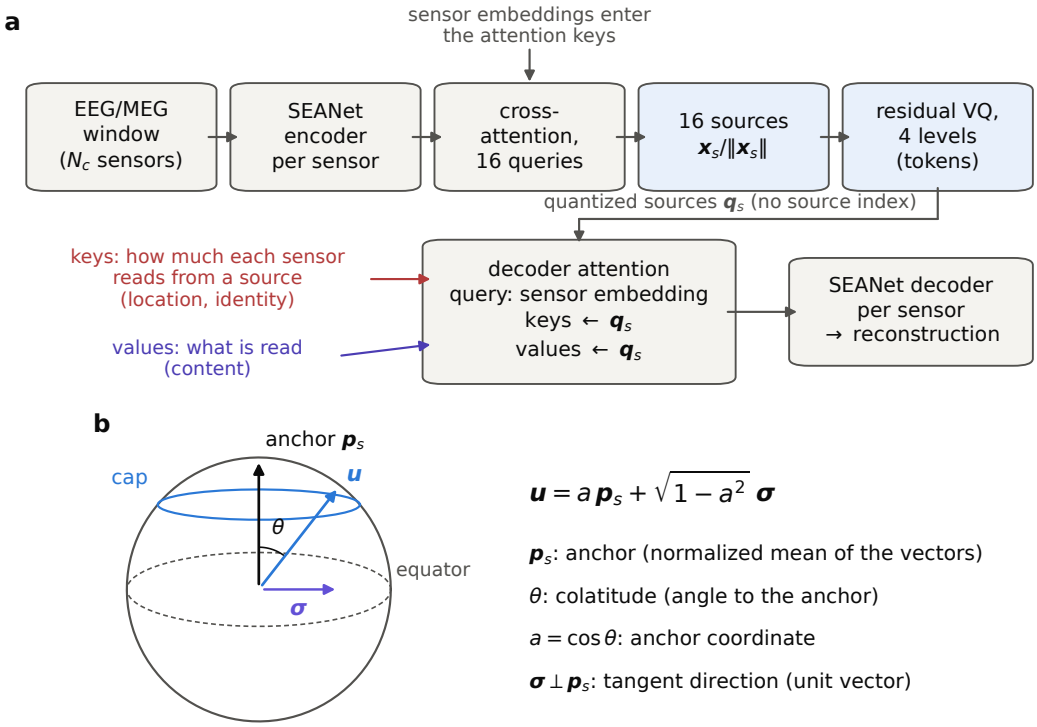


Figure 1: (a) The BrainOmni tokenizer. Every sensor's window is encoded separately; 16 learned queries pool the sensors into 16 latent sources (the sensor embeddings enter the keys of this cross-attention), whose unit vectors are quantized by a four-level residual vector quantizer. The decoder lets each sensor's embedding attend over the quantized sources: the attention keys, computed from the quantized vectors, determine how much each sensor reads from each source, and the values determine what is read. The decoder receives no source index, so the sources can be told apart only through their quantized vectors. (b) Hyperspherical coordinates of the unit vectors $\boldsymbol{u}$ of one source: the anchor $\boldsymbol{p}_s$ (the normalized mean of the vectors), the colatitude θ (the angle to the anchor), the anchor coordinate $a = \cos\theta$ and the tangent direction $\boldsymbol{\sigma}$; the vectors of a source lie in a cap around its anchor.

2025). (The pretraining data were processed in the same way, except that they were notch filtered at both 50 and 60 Hz, resampled before band-pass filtering, had bad EEG channels interpolated, and had the positions of all MEG sensors normalized, whereas the downstream scripts normalize only those of magnetometers.) Every sensor c has a six-dimensional descriptor (its position, centered and scaled per modality, and its orientation, which is zero for EEG) and a type (EEG, magnetometer or gradiometer), which a small network maps to a sensor embedding $\boldsymbol{e}_c \in \mathbb{R}^{256}$. The tokenizer processes 2-s windows (512 samples) independently; a 10-s segment contains five windows.

Encoder and latent sources. A convolutional encoder with a residual two-layer long short-term memory (LSTM) network (SEANet, total stride 64; Zeghidour et al., 2022; Défossez et al., 2023) maps the window of each sensor to $F = 8$ frames of 256-dimensional features $\boldsymbol{h}_{c,f}$. For every frame, 16 learned query vectors $\boldsymbol{n}_1, \ldots, \boldsymbol{n}_{16}$ attend over the sensors through four-head cross-attention whose keys are computed from $\boldsymbol{h}_{c,f} + \boldsymbol{e}_c$ and whose values are computed from $\boldsymbol{h}_{c,f}$,

$$\begin{aligned} \boldsymbol{x}_{s,f} &= W_o \,\mathrm{Attn}\big(\boldsymbol{n}_s;\, \{\boldsymbol{k}_{c,f},\, \boldsymbol{z}_{c,f}\}_{c=1}^{N_c}\big) + \boldsymbol{b}_o \in \mathbb{R}^{256}, \qquad s = 1, \ldots, 16, \\ \boldsymbol{k}_{c,f} &= W_k(\boldsymbol{h}_{c,f} + \boldsymbol{e}_c) + \boldsymbol{b}_k, \qquad \boldsymbol{z}_{c,f} = W_v \boldsymbol{h}_{c,f} + \boldsymbol{b}_v. \end{aligned} \tag{1}$$

We call $\boldsymbol{x}_{s,f}$ the vector of *source* s at frame f. Because the number of queries is fixed, the representation has the same shape for every montage and modality.

Quantizer. Each source vector is normalized, $\boldsymbol{u} = \boldsymbol{x}/\|\boldsymbol{x}\|$, and quantized by a four-level residual vector quantizer: level ℓ replaces the current residual by its nearest codeword $\boldsymbol{c}_\ell$ in a codebook of 512 entries and passes on the difference, so that the quantized vector is $\boldsymbol{q} = \sum_{\ell=1}^{4} \boldsymbol{c}_\ell$. The four codebooks

are shared by all sources and were learned with exponential-moving-average k-means updates. The norm $\|\boldsymbol{x}\|$ is discarded: the tokenizer passes on only the direction of each source vector, through its four code indices.

Decoder and objective. For every frame, the decoder lets the embedding of each sensor attend over the 16 quantized sources (four-head attention with the sensor embedding as query, and keys and values that are affine functions of the quantized source vectors) and maps the result back to that sensor's waveform with a convolutional decoder that mirrors the encoder. The decoder receives no source index or source embedding, so it can tell the sources apart only through their quantized vectors; and it reconstructs every sensor from that sensor's embedding and the quantized sources alone, independently of the other sensors. The tokenizer was trained to reconstruct all sensors of a window, z-scored per sensor and window, while its encoder saw a random 75% of the sensors with additive Gaussian noise. The loss is the sum of the mean absolute error, $\exp(-r)$ with r the mean Pearson correlation between reconstructed and target waveforms, the mean absolute errors of the spectral amplitude and (weighted by 0.5) of the spectral phase, and the commitment term of the quantizer (Xiao et al., 2025). The tokenizer has 5.05 million trainable parameters.

Tokenizer instances. We analyze (i) the released BrainTokenizer checkpoint; (ii) the tokenizer stored inside the released BrainOmni foundation-model checkpoints (identical in the *tiny* and *base* models), whose encoder and decoder equal those of (i) up to bfloat16 rounding but whose codebooks differ; (iii) four tokenizers trained from scratch with the released architecture and objective (Section 4.5), two of them with a learned source embedding added

to the decoder's input; and, as a control for geometry only, (iv) untrained tokenizers with random weights. Unless stated otherwise, results refer to (i).

4. Methods

4.1. Cohorts and preprocessing

We used four public OpenNeuro datasets (Markiewicz et al., 2021) (Table 1): resting-state EEG from patients with Parkinson's disease off medication and healthy controls recorded at UC San Diego (ds002778; Rockhill et al., 2021; Jackson et al., 2019; Swann et al., 2015; George et al., 2013) (*PD-UCSD*); eyes-open resting-state EEG from patients with Parkinson's disease and controls recorded at the University of Iowa (ds004584; Singh et al., 2023a,b; Anjum et al., 2024) (*PD-Iowa*); eyes-closed resting-state EEG from patients with Alzheimer's disease, frontotemporal dementia and cognitively normal controls (ds004504; Miltiadous et al., 2026a,b, 2023a,b) (*Dementia*); and resting-state MEG from adolescents with major depressive disorder and healthy volunteers (ds003568; Liuzzi et al., 2023, 2022) (*MDD-MEG*). All subjects with a usable recording were included; PD-UCSD patients contributed their off-medication session, and one PD-Iowa recording was excluded because its median channel standard deviation exceeded 500 µV. For MDD-MEG, the first 190 s of each resting recording (310–600 s long) were read and filtered, and its first 18 complete segments (the first 180 s) were analyzed.

Preprocessing followed the BrainOmni pipeline for downstream data: notch filtering at the mains frequency, 0.1–96 Hz band-pass filtering, resampling to 256 Hz, non-overlapping 10-s segments and per-segment normalization as described in Section 3. No artifact rejection was applied: segments overlapping

Table 1: Cohorts. All recordings are resting state (PD-Iowa eyes open, Dementia eyes closed); segments are non-overlapping 10-s sections of a recording (five 2-s tokenizer windows each); group sizes count the subjects analyzed. HC: healthy controls; PD: Parkinson's disease; CN: cognitively normal; AD: Alzheimer's disease; FTD: frontotemporal dementia; HV: healthy volunteers; MDD: major depressive disorder.

cohort	**accession**	**modality**	**sensors**	**subjects**	**segments**	**groups**
PD-UCSD	ds002778	EEG	32	31	591	HC 16, PD 15
PD-Iowa	ds004584	EEG	60	148	2,167	HC 49, PD 99
Dementia	ds004504	EEG	19	88	7,013	CN 29, AD 36, FTD 23
MDD-MEG	ds003568	MEG	269	49	882	HV 20, MDD 29

intervals annotated as bad or as recording boundaries would have been discarded, but the only such annotation was the boundary event at the start of every PD-Iowa recording, so the first 10 s of the PD-Iowa recordings were not used. EEG sensors were matched to the standard 10–05 or BioSemi template positions and restricted to the channels shared by all recordings of a cohort; MEG used 269 of the 272 axial gradiometers (three excluded by the fixed list of the BrainOmni MEG pipeline) after third-order gradient compensation. For MEG we departed from the downstream pipeline in two respects: segments were 10 s instead of 30 s long, as for EEG, and the positions of all MEG sensors were normalized, as in the tokenizer's pretraining. Throughout, *segment* refers to a retained 10-s segment and *window* to one of its five 2-s tokenizer inputs. Preprocessing was performed with MNE-Python (Gramfort et al., 2013) on data organized in the Brain Imaging Data Structure (BIDS) (Pernet et al., 2019; Niso et al., 2018; Appelhoff et al., 2019).

4.2. Per-source unit vectors and anchor geometry

A segment yields $5 \times 8 = 40$ unit vectors per source (five windows of eight frames). For source s of a cohort, let $\boldsymbol{u}$ denote a random unit vector of that source, drawn uniformly from its vectors in all segments used for the estimate.

We define the directional mean, its *mean resultant length* and the *anchor* (Figure 1b),

$$\boldsymbol{\mu}_s = \mathbb{E}[\boldsymbol{u}], \qquad \bar{R}_s = \|\boldsymbol{\mu}_s\| \in [0,1], \qquad \boldsymbol{p}_s = \boldsymbol{\mu}_s / \bar{R}_s, \tag{2}$$

which are the standard location and concentration parameters of directional data (Mardia and Jupp, 2000). Every unit vector has hyperspherical coordinates relative to the anchor,

$$\boldsymbol{u} = a\,\boldsymbol{p}_s + \sqrt{1-a^2}\,\boldsymbol{\sigma}, \qquad a = \langle \boldsymbol{u}, \boldsymbol{p}_s \rangle = \cos\theta, \qquad \boldsymbol{\sigma} \perp \boldsymbol{p}_s,\ \|\boldsymbol{\sigma}\| = 1, \tag{3}$$

consisting of the *anchor coordinate* a (equivalently the *colatitude* θ, the angle between $\boldsymbol{u}$ and the anchor) and the *tangent direction* $\boldsymbol{\sigma}$; we write $\boldsymbol{t} = \boldsymbol{u} - a\boldsymbol{p}_s = \sqrt{1-a^2}\,\boldsymbol{\sigma}$ for the tangent part. Note that $\mathbb{E}[a] = \bar{R}_s$. Three kinds of principal directions are distinguished. The *uncentered principal directions* are the eigenvectors of the second-moment matrix $M_s = \mathbb{E}[\boldsymbol{u}\boldsymbol{u}^\top]$; the *centered principal components* (PCs) are those of the covariance $\Sigma_s = M_s - \boldsymbol{\mu}_s\boldsymbol{\mu}_s^\top$; and the *tangent principal axes* are the eigenvectors of the covariance of the tangent direction $\boldsymbol{\sigma}$, which lie in the hyperplane orthogonal to the anchor. The following proposition relates the anchor to the leading uncentered direction $\boldsymbol{v}_1$ of M_s; it combines elementary identities with the residual form of the Davis–Kahan $\sin\Theta$ theorem (Davis and Kahan, 1970) and Weyl's inequality (proof in Appendix A). We write $\lambda_1 \geq \lambda_2 \geq \dots$ for the eigenvalues of M_s and $\boldsymbol{\gamma}_s = \mathrm{Cov}(a, \boldsymbol{t}) = \mathbb{E}[a\boldsymbol{t}]$ for the cross-covariance between the anchor coordinate and the tangent part.

Proposition 1. *For unit vectors with $\bar{R}_s > 0$:*

(i) $M_s\boldsymbol{p}_s = \mathbb{E}[a^2]\,\boldsymbol{p}_s + \boldsymbol{\gamma}_s$ with $\boldsymbol{\gamma}_s \perp \boldsymbol{p}_s$; in particular, $\boldsymbol{p}_s^\top M_s \boldsymbol{p}_s = \mathbb{E}[a^2]$, and $\boldsymbol{p}_s$ is an eigenvector of M_s if and only if $\boldsymbol{\gamma}_s = \boldsymbol{0}$.

(ii) If $\mathbb{E}[a^2] > \lambda_2$, then, for $\boldsymbol{v}_1$ oriented so that $\langle \boldsymbol{v}_1, \boldsymbol{p}_s \rangle \geq 0$,

$$\sin \angle(\boldsymbol{v}_1, \boldsymbol{p}_s) \;\leq\; \frac{\|\boldsymbol{\gamma}_s\|}{\mathbb{E}[a^2] - \lambda_2}\,. \tag{4}$$

(iii) $\lambda_2 \leq \|\Sigma_s\|_{\mathrm{op}}$, so (ii) also holds with λ_2 replaced by $\|\Sigma_s\|_{\mathrm{op}}$ whenever $\mathbb{E}[a^2] > \|\Sigma_s\|_{\mathrm{op}}$.

(iv) $\|\boldsymbol{\gamma}_s\| \leq \sqrt{\mathrm{Var}(a)\,\|\Sigma_s\|_{\mathrm{op}}}$, so the right-hand side of (4) is at most $\sqrt{\mathrm{Var}(a)\,\|\Sigma_s\|_{\mathrm{op}}}\,/\,(\mathbb{E}[a^2] - \lambda_2)$.

(v) $\operatorname{tr}\Sigma_s = 1 - \bar{R}_s^2$ and $\Sigma_s \boldsymbol{p}_s = \mathrm{Var}(a)\,\boldsymbol{p}_s + \boldsymbol{\gamma}_s$; in particular, the variance along the anchor is $\boldsymbol{p}_s^\top \Sigma_s \boldsymbol{p}_s = \mathrm{Var}(a) = \mathbb{E}[a^2] - \bar{R}_s^2$.

By (i), the anchor is an exact eigenvector of M_s exactly when the anchor coordinate is uncorrelated with the tangent part, and by (ii) it stays close to the leading eigenvector as long as the norm of their cross-covariance is small compared with the gap between $\mathbb{E}[a^2]$ and the second eigenvalue. By (iv), this is guaranteed whenever the anchor coordinate varies little, whatever its correlation with the tangent part: concentration in colatitude together with an eigengap suffices. The classical perturbation bound $\sin \angle(\boldsymbol{v}_1, \boldsymbol{p}_s) \leq 2\|\Sigma_s\|_{\mathrm{op}}/\bar{R}_s^2$ (Davis and Kahan, 1970; Yu et al., 2015), which treats all of Σ_s as a perturbation of $\boldsymbol{\mu}_s \boldsymbol{\mu}_s^\top$, is much weaker because it ignores that most of the variation is tangential. By (i) and (v), the anchor satisfies the eigenvalue equations of both second moments up to the same residual $\boldsymbol{\gamma}_s$: that of M_s with the Rayleigh quotient $\mathbb{E}[a^2]$, which under the condition of (ii) lies between λ_2 and λ_1, and that of Σ_s with the Rayleigh quotient $\mathrm{Var}(a)$, which may be a small share of $\operatorname{tr}\Sigma_s$. The anchor can therefore be the leading uncentered direction without being a leading centered one. Whether it is close to a single eigenvector depends on the distance from the corresponding eigenvalue

to the others: for M_s this is the eigengap of (ii); applied to Σ_s, the same argument guarantees closeness to a single centered PC only if $\|\boldsymbol{\gamma}_s\|$ is small compared with the distance from $\mathrm{Var}(a)$ to the neighboring eigenvalues of Σ_s, and otherwise the anchor can be spread over several centered PCs of similar variance.

For every source we report $\bar{R}_s$; the mean colatitude; $\cos\angle(\boldsymbol{v}_1, \boldsymbol{p}_s)$ together with the bound (4), its versions (iii) and (iv) and the classical bound; the absolute cosine between the anchor and each of the ten leading centered PCs, and the largest absolute cosine between the anchor and any of the 64 leading centered PCs together with the rank of that PC; the squared norm of the projection of the anchor onto the span of the k leading centered PCs ($k = 10, 16, 64$); the share of $\operatorname{tr}\Sigma_s$ along the anchor and its rank among the eigenvalues of Σ_s; the participation ratio $(\operatorname{tr}\Sigma_s)^2/\operatorname{tr}(\Sigma_s^2)$ as an effective dimensionality of the variation; and the share of the variance of the tangent direction captured by its 16 and 64 leading tangent axes. For the anchors of all 16 sources together, we report how many directions they span, as the participation ratio of the eigenvalues of their cosine matrix $G_{ss'} = \langle \boldsymbol{p}_s, \boldsymbol{p}_{s'} \rangle$ (the cumulative shares of these eigenvalues are the fractions of the anchors' squared norm in their leading principal directions), and the smallest angle between two anchors. When the two leading eigenvalues of G hold at least 99% of their sum, we also report the gaps between consecutive anchors along the circle, the differences between the polar angles of the anchors projected onto the plane of the two leading eigenvectors of G. Geometry was estimated from the second to fourth segment of every recording (120 vectors per subject and source; the first segment was excluded to avoid filter transients). Reliability

was assessed by split halves over subjects (30 random splits), by the agreement of each subject's anchor with the anchor of the remaining subjects, and by the agreement of anchors estimated on different cohorts. Intervals are 95% subject-cluster bootstrap intervals (200 resamples).

4.3. Interventions

To measure which aspects of a source the decoder relies on, we edit the unit vectors of one source in a segment, re-run the frozen quantizer and decoder, and record the change in reconstruction error (mean squared error, MSE) relative to the unedited reconstruction,

$$\Delta\mathrm{MSE} = \mathrm{MSE}\big(\hat{X}_{\mathrm{edit}}, X\big) - \mathrm{MSE}\big(\hat{X}, X\big), \tag{5}$$

where X is the tokenizer's training target (each sensor and window z-scored) and the error is averaged over sensors, windows and samples; we report ΔMSE mostly in percent of the unedited error of the segment, and compute the change in mean absolute error (ΔMAE) and in one minus the mean Pearson correlation between reconstructed and target waveforms ($\Delta(1-r)$) in the same way. Because the quantizer normalizes its input, edits that move vectors off the unit sphere—such as scaling or averaging the coordinate along a direction—would be partly undone by the normalization before they reached the codebooks: halving the anchor component of every vector without renormalizing would, after the quantizer's normalization, have achieved only 70–89% of the intended change in the anchor coordinate (median over the vectors of a source), and setting it to its mean would have left 7–25% of the variance of the anchor coordinate. All edits are therefore defined in the hyperspherical coordinates of Equation (3) and map unit vectors to

unit vectors (Table 2). They separate three aspects of a source. *Location* edits change the anchor coordinate and keep the tangent direction: setting it to zero (*zero-anchor*: moving every vector to the anchor's equator; this equals projection, or zero, ablation of the anchor followed by the quantizer's normalization), halving it, replacing it by its mean or by the coordinate of a randomly permuted vector of the same segment, and shifting the colatitude by $\pm 10^\circ$ or $\pm 20^\circ$ (vectors closer to the anchor than a negative shift are moved onto it). *Content* edits keep the anchor coordinate and change the tangent direction: replacing its components on the k leading tangent axes by those of the vector at the same position in a segment from another subject (resample ablation; $k = 1, 4, 16, 64$), by those of the mean of the tangent directions, $\mathbb{E}[\boldsymbol{\sigma}]$, which is not normalized and short, so that this nearly zeroes these components before renormalization (mean ablation; $k = 1, 4$), or replacing the whole tangent direction. As controls, the same resample ablation was applied to k random orthonormal directions inside the span of the 64 leading tangent axes, which contains almost all of the tangent variance (the *support*; $k = 1, 4, 16$), and to 16 random directions of the whole tangent space (orthogonal to the anchor, with on average a quarter of their squared norm in the support); and the tangent direction was rotated by a fixed angle or by a per-vector angle chosen such that the geodesic displacement equals that of zero-anchor (*matched rotation*). A matched rotation keeps the colatitude and moves each vector by the same angle as zero-anchor, around the anchor instead of towards its equator, which separates the effect of the direction of an edit from that of its size; for vectors within 30° of the anchor (a median of 0.4% and at most 5% of the vectors of a source), no such angle exists and the

rotation is by 180°. The *source-identity* edit rotates all vectors of a source in the plane spanned by its anchor and the anchor of another source so that its anchor is mapped onto the other one (*anchor swap*), which preserves every angle within the source. As an extreme, *collapse* replaces every vector by the anchor, which removes all content and sets the colatitude to zero while keeping the anchor, and *source resample* replaces all vectors of a source by those of a segment from another subject, which gives a reference level for the damage caused by replacing a source with realistic content.

To locate these effects in the decoder, we applied zero-anchor and the anchor swap to only one of the two pathways through which the quantized sources enter the decoder's attention: the input of its keys (which determine how much each sensor reads from each source) or that of its values (what is read), the other pathway receiving the unedited quantized vectors. A second run on the same segments applied the same split to zero-anchor, to the anchor swap, to the resample of the whole tangent direction and to the resample of the four leading tangent axes; the key and value shares plotted for these four edits (Section 5.5) and the routing of the content edits reported there come from this run, all other zero-anchor and swap values from the first (the two runs gave the same zero-anchor shares and swap shares within 0.01). For zero-anchor we also recorded the attention weight that the sensors assign to the edited source, averaged over sensors, heads and frames.

Anchors, mean anchor coordinates, mean tangent directions and tangent axes were cross-fitted: subjects were split at random into two halves, and the segments of each half were edited with statistics estimated on the other. Edits were applied to every source of two segments per subject in PD-UCSD

Table 2: Edits applied to the unit vectors of one source, in the hyperspherical coordinates of Equation (3) (anchor $\boldsymbol{p}$, anchor coordinate $a = \cos\theta$, tangent direction $\boldsymbol{\sigma} \perp \boldsymbol{p}$). Every edit maps unit vectors to unit vectors. $\boldsymbol{\sigma}'$ and $\boldsymbol{u}'$ denote the tangent direction and the unit vector at the same position of a segment from another subject; the *support* is the span of the 64 leading tangent axes. Statistics ($\boldsymbol{p}$, $\bar{a}$, tangent axes, mean of the tangent directions) are cross-fitted (Section 4.3). Routing: zero-anchor, the anchor swap, tangent resample and tangent-axis resample (4) were also applied only to the input of the decoder's attention keys, or only to that of its values.

edit	**definition**	**changes**
Location: $\boldsymbol{\sigma}$ kept		
zero-anchor	$a \leftarrow 0$ ($\theta \leftarrow 90°$)	colatitude
half-anchor	$a \leftarrow a/2$	colatitude
mean-anchor	$a \leftarrow \bar{a}$ (mean anchor coordinate)	colatitude variation
resample-anchor	$a \leftarrow$ coordinates randomly permuted among the vectors of the segment	pairing of a and $\boldsymbol{\sigma}$
colatitude shift	$\theta \leftarrow \min\{\max(\theta + \delta, 0°), 180°\}$, $\delta \in \{\pm 10°, \pm 20°\}$ (clipped at the anchor)	colatitude
Content: a kept		
tangent-axis resample (k)	components of $\boldsymbol{\sigma}$ on the k leading tangent axes $\leftarrow$ those of $\boldsymbol{\sigma}'$; renormalized	content in the leading axes
tangent-axis mean (k)	as above with the mean $\mathbb{E}[\boldsymbol{\sigma}]$ of the tangent directions (not normalized) instead of $\boldsymbol{\sigma}'$	content in the leading axes
random-support resample (k)	as tangent-axis resample, k random orthonormal directions of the support	content in a random subspace
random resample (16)	as above, 16 random directions of the whole tangent space ($\perp \boldsymbol{p}$)	content in a random subspace
tangent resample	$\boldsymbol{\sigma} \leftarrow \boldsymbol{\sigma}'$	all content
matched rotation	$\boldsymbol{\sigma} \leftarrow \cos\psi\,\boldsymbol{\sigma} + \sin\psi\,\boldsymbol{\xi}$, $\boldsymbol{\xi} \perp \boldsymbol{\sigma}$ random in the support (or in $\boldsymbol{p}^{\perp}$), ψ giving zero-anchor's displacement ($\psi = 180°$ within 30° of $\boldsymbol{p}$)	content, at equal displacement
rotation (ψ)	as above with $\psi = 30°, 60°, 90°$, $\boldsymbol{\xi}$ in the support	content
Source identity and whole source		
anchor swap	rotation in $\mathrm{span}\{\boldsymbol{p}_s, \boldsymbol{p}_{s'}\}$ taking $\boldsymbol{p}_s$ to the anchor $\boldsymbol{p}_{s'}$ of another source	which anchor
collapse	$\boldsymbol{u} \leftarrow \boldsymbol{p}$	all content and colatitude
source resample	$\boldsymbol{u} \leftarrow \boldsymbol{u}'$	everything

and MDD-MEG (the third and the seventh segment of the recording) and of one segment per subject in the larger PD-Iowa and Dementia cohorts (the third). Segments were processed in groups of 16 (8 for MDD-MEG), each containing segments of at least two subjects, from which the donor segments of the resample edits were drawn (one random assignment of every segment to a segment of another subject per group, shared by all sources and all resample edits, so that comparisons between edits are paired). For every source and group, three partner sources were drawn for the anchor swap (the split into keys and values used the first), two random support subspaces of each dimension, one random 16-dimensional subspace of the tangent space, and one random rotation direction for every vector. For the 60- and 269-channel cohorts the reconstruction was evaluated on a fixed random subset of 32 sensors, which is exact for the sampled sensors because the decoder reconstructs every sensor independently of the others (Section 3); the encoder always saw all sensors. Results were averaged per subject (over segments and draws); for every source we report the mean over subjects, with a 95% subject-cluster bootstrap interval (2000 resamples), and differences and ratios between edits are computed from the subject means with the same resamples (paired bootstrap). We call an effect *positive for a source* when the interval of its subject-mean ΔMSE lies above zero; these counts, and the differences and ratios between edits, are computed from ΔMSE itself; values given in percent of the unedited error use the change relative to the unedited error of each segment, and shares between edits are ratios of ΔMSE.

4.4. Token analysis

For every RVQ level ℓ and cohort we computed, over the vectors of all segments, the mutual information $I(S; C_\ell)$ between the source index S and the code C_ℓ in bits (with 16 equally frequent sources, $H(S) = 4$ bits), its share of the entropy of the code, the accuracy of identifying the source from the code alone (each code assigned to the source that uses it most often), and the perplexity of the code usage of every source. For level 1 we computed the usage-weighted cosine between the codewords a source selects and its anchor. We then re-quantized the vectors used for the geometry after editing them as in Table 2 (with statistics estimated on the same vectors and donor tangent directions taken from a random segment of another subject), and recorded the fraction of vectors whose level-1 code changed, the fraction of these changes that went to a code used most often by another source, and, among vectors whose level-1 code did not change, the fraction whose level-2 code changed.

4.5. Independently trained tokenizers

To test whether the findings depend on one trained checkpoint, we trained tokenizers from scratch with the released architecture (16 sources, four codebooks of 512 entries) and objective (reconstruction of all sensors from 75% of them, with the loss of Section 3; input noise of standard deviation 0.1, commitment term 0.25 times the mean squared quantization error averaged over levels, codebooks updated by the moving-average rule with decay 0.99 and initialized by k-means on the first batch, codewords whose moving-average usage count fell below 2 re-initialized) and with the released optimizer (AdamW with $\beta = (0.5, 0.9)$, $\epsilon = 10^{-5}$, learning rate 2×10^{-4}, weight decay 0.01 except

for normalization weights and the sensor-type embedding; cosine decay of the learning rate to 0.05 times its initial value after a warm-up over 10% of the steps). The subjects of every cohort were split, stratified by group, into a training set (75%) and a held-out set (25%; 8, 37, 22 and 12 subjects in PD-UCSD, PD-Iowa, Dementia and MDD-MEG); the same split was used for all tokenizers we trained, and every analysis of them used held-out subjects only. Each training step drew one cohort at random and a batch of eight ten-second segments, with replacement, from its training subjects, restricted to one random subset of 16 EEG or 32 MEG sensors shared by the segments of the step; because the decoder reconstructs every sensor independently of the others (Section 3), a sensor subset is a valid training input. Training ran for 3,000 steps. Two tokenizers (*replicas*; replica k uses seed k) differ only in their random seed. Two further tokenizers use the seeds of the replicas, so that their initial weights (apart from the added embedding) and their sequences of training batches equal those of the replicas, and add a learned embedding $\boldsymbol{\eta}_s \in \mathbb{R}^{256}$ of each source, initialized from a truncated normal distribution with variance $1/256$, to the quantized source vectors before the decoder's key and value projections (*source-embedding variants*; variant k uses the seed of replica k). The variants give the decoder a second way to identify the sources, independent of their direction, and therefore test whether the role of the anchors depends on the decoder having no other access to source identity, and each variant is compared with the replica of the same seed.

We made three changes to the published recipe. First, with the original initialization all sources coincided (Section 5.1) and, in a pilot run of 3,000 steps with the recipe otherwise as for the replicas, did not separate (held-out

PD-UCSD: mean resultant length 0.93–0.97, pairwise cosine between the anchors ≥ 0.99; training-batch correlation between reconstructed and target waveforms 0.12 at the end of training, against 0.68 for replica 1); we therefore initialized the convolution biases of the encoder and decoder to zero and drew the weights of the sensor-position network and of the encoder's attention keys with standard deviation 0.2 instead of 0.02. Second, the codebook implementation re-initializes unused codewords from the current batch, but the replacement is overwritten by the following moving-average update; we made the replacement effective. Third, for numerical stability, we clipped the gradient norm at 1, trained in single precision instead of bfloat16, and used a linear instead of the released logarithmic warm-up. The training data comprised 22 h of recordings, a small fraction of the corpus of the released tokenizer, and the batch was eight segments on sensor subsets instead of the 512 full segments of the released training. The replicas therefore test whether the geometry and its causal roles emerge from the architecture and objective under a modified initialization and a much smaller training budget, not whether the released tokenizer's reconstruction quality can be reached. We report their reconstruction error on held-out subjects next to that of the released tokenizer, and the geometry of the first replica at initialization as an untrained control. Geometry, tokens and interventions were analyzed as for the released tokenizer, on the held-out subjects, except that intervention segments were processed in groups of eight in every cohort and that one random support subspace of each dimension and one random 16-dimensional subspace were drawn per group and source; the held-out sets are small (8 to 37 subjects), so the subject-cluster intervals of these tokenizers are approximate.

Because the caps of the replicas are narrow (Section 5.7; a median of 95% of the vectors of a source lie within 30° of the anchor), the matched rotation is the 180° fallback for most of their vectors and moves them less than zero-anchor, shifts of the colatitude towards the anchor move many vectors onto it, and halving the anchor component off the sphere would have achieved only 10–36% of the intended change.

4.6. Diagnostic read-out

For every subject, per-segment quantities were averaged over all of the subject's segments. The primary representation was the vector of the 16 anchor coordinates $\langle \bar{\boldsymbol{u}}_s, \boldsymbol{p}_s \rangle$, where $\bar{\boldsymbol{u}}_s$ is the subject's mean unit vector of source s (*anchor-16*). Secondary representations were the full per-source representation $(\bar{\boldsymbol{u}}_1, \ldots, \bar{\boldsymbol{u}}_{16}) \in \mathbb{R}^{4096}$ (*full*); the same with the anchor coordinates removed (*tangent*); the mean over sources (*pooled*, 256-D); the per-source means of the quantized vectors $\boldsymbol{q}$, the tokenizer's output (*quantized*, 4096-D), and their anchor coordinates (*quantized anchor-16*); the mean resultant lengths of the vectors of each source within a segment (*concentration-16*); the mean norms before normalization, which the quantizer discards (*norm-16*); and log band power in five bands (1–4, 4–8, 8–13, 13–30, 30–45 Hz) per sensor, computed from the preprocessed segments (average-referenced and scaled by their global standard deviation, so that it measures relative power). Anchors were re-estimated on the training subjects of every cross-validation fold.

Classification used L2-regularized logistic regression on standardized features (for high-dimensional representations the standardized training data were first rotated onto their non-degenerate principal axes, which is lossless for an L2 penalty). The inverse regularization strength was chosen from 13

log-spaced values between 10^{-4} and 10^{2} by the AUC of an inner stratified five-fold cross-validation on the training subjects, and the model was then refitted on all training subjects. Performance is the AUC in stratified five-fold cross-validation repeated 20 times: in every repetition the out-of-fold scores of all subjects were pooled over the five folds and one AUC was computed, and we report the mean over repetitions. Its interval is a 95% stratified bootstrap over subjects of the same statistic with the out-of-fold scores held fixed (2000 resamples); it describes the uncertainty of evaluating the fitted classifiers on these subjects, not that of the cross-validated AUC, and inference rests on permutation tests. P-values for AUCs above chance were obtained by permuting the diagnostic labels across subjects and re-running the entire cross-validation, $p = (1 + \#\{\text{null} \geq \text{observed}\})/(1 + n_{\text{perm}})$; because nested selection of the regularization strength inside every permutation is costly, the permutation test uses the same pipeline with the inverse regularization strength fixed at 1 and two repetitions, for the observed and the permuted labels alike, which keeps the test exact for that pipeline (its statistic differed from the reported AUC by at most 0.12). The primary hypothesis (anchor-16 above chance) was tested with 1000 permutations per contrast and Holm-corrected across the five contrasts (Holm, 1979); secondary representations were tested with 500 permutations and Benjamini–Hochberg-corrected within each contrast (Benjamini and Hochberg, 1995). Single anchor coordinates were read out with the inverse regularization strength fixed at 1 and tested with a single-step maximum-statistic permutation test across the 16 sources of a contrast (1000 permutations; Westfall and Young, 1993); in addition, a nested procedure selected, inside every training fold, the anchor coordinate

whose AUC on the training subjects was farthest from 0.5 and evaluated it out of fold (*best single anchor*; the selection was repeated inside every one of 1000 permutations). As a null for anchor-16, the identical procedure was applied to 16 random unit directions (one per source; 100 draws), with the Monte Carlo p-value $(1 + \#\{\text{random} \geq \text{observed}\})/(1 + n_{\text{rand}})$. Differences between representations were assessed on the same subjects with a paired stratified bootstrap of the difference of their AUCs (2000 resamples, out-of-fold scores held fixed), for every representation against band power and against anchor-16; for this comparison the out-of-fold scores were recomputed with the same folds and seeds, which reproduced the reported AUCs to within 0.001. The p-value of a paired difference is two-sided: twice the smaller of the fractions of bootstrap resamples in which the difference is at most or at least zero. Like the intervals, these differences hold the out-of-fold scores fixed and ignore the variability of model fitting, so they are descriptive rather than tests; because every bootstrap resample averages the AUC over the 20 repetitions, their intervals narrow as the out-of-fold predictions differ more between repetitions. To assess confounding by age and sex, we evaluated a classifier on age and sex alone and repeated the analysis of anchor-16, the full representation and band power with both regressed out of the features (least squares fitted on the training subjects of every fold). These p-values, which are not corrected for multiple testing, come from permuting the labels while age and sex stay with the features; they test the joint null that the labels are independent of the features and of the covariates, and therefore do not isolate information beyond age and sex.

To test transfer across sites and montages, a classifier was trained on all

subjects of one Parkinson's cohort and applied to all subjects of the other, which is possible because the source representations have the same dimensionality for every montage (band power was restricted to the 29 electrodes shared by the two montages). Anchors were estimated on the training cohort, and the regularization strength was chosen by stratified five-fold cross-validation on it. Features were standardized with the training cohort's statistics or, in a second analysis, within each cohort (a label-free harmonization). For every representation, the AUC of the fitted classifier on the test cohort is given with a stratified bootstrap interval over the test subjects (scores held fixed). Its p-value was obtained by permuting the labels of the test cohort with the classifier fixed (10,000 permutations), which is exact under the null hypothesis that the transferred scores are unrelated to the test labels; as within cohorts, the p-values of anchor-16 were Holm-corrected across the two directions and those of the secondary representations Benjamini–Hochberg-corrected within each direction. Each secondary representation was compared with anchor-16 by a paired stratified bootstrap of the difference of their AUCs on the test subjects (Holm-corrected across the eight comparisons of a direction), and every representation with the AUCs of 1,000 random 16-direction representations evaluated like anchor-16; for anchor-16 this gives a Monte Carlo p-value as within cohorts, Holm-corrected across the two directions and, as a check, across both directions and both scalings.

4.7. Software and reproducibility

All analyses used PyTorch 2.11, scikit-learn 1.8 and MNE-Python 1.12 on a single Apple M2 Pro computer. The analysis code was developed and the analyses were run with the assistance of Claude Opus 4.7 and Claude

Opus 5.5 (Anthropic; see the Declaration of generative AI and AI-assisted technologies). For every export and intervention run, GPU (Metal) results were checked against CPU results on its first four segments: the unit vectors differed by at most 2.2×10^{-6}, and the codes were identical for the released and embedded tokenizers and agreed for at least 99.92% of the vectors for the tokenizers we trained (the untrained controls, whose codebooks are initialized from the first batch, were used for geometry only). Code, the exact segment and subject lists, and the intermediate results needed to regenerate every number, table and figure are available as described in the Data availability statement.

5. Results

5.1. Each source occupies a cap around a tokenizer-specific anchor

Table 3 and Figure 2 summarize the geometry of the per-source unit vectors. In every cohort, the vectors of every source were concentrated around their anchor (Figure 2a): mean resultant lengths ranged from 0.49 to 0.73 in the EEG cohorts and from 0.57 to 0.71 in MEG, and the mean colatitude ranged from 43° to 60°, whereas two independent random directions in 256 dimensions are 90° apart to within a few degrees. Each source therefore occupies a cap of the hypersphere around its anchor. The anchors were estimated precisely: anchors computed from random halves of the subjects agreed to cosine ≥ 0.984, and the anchor of a single subject agreed with that of the remaining subjects to a median cosine of 0.973–0.993 across cohorts (5th percentiles 0.76–0.99). Anchors of different sources were distinct but not orthogonal (pairwise cosines between -0.72 and 0.80, median absolute

cosine 0.35, compared with about 0.04 for random directions). Together, the 16 anchors lay close to a four-dimensional subspace: their four leading principal directions captured at least 98.6% of their summed squared norm (participation ratio 3.9–4.0), and the smallest angle between two anchors was 37–39°.

The anchors of a source estimated on different cohorts were closely aligned (Figure 2c): per-source cosines were at least 0.976 among the three EEG cohorts, which differ in site, montage (19 to 60 channels), population and recording condition (angles of at most 13°), and at least 0.93 between the EEG cohorts and MEG (at most 21°), compared with at least 37° between the anchors of different sources. The anchors are therefore largely fixed by the tokenizer; the recording modality shifts them by up to 21°, less than the separation between the anchors of different sources. They are a product of training: with the released initialization and random weights, the vectors of all 16 sources pointed in one common direction (PD-UCSD; mean resultant length 1.00, pairwise cosine between the anchors ≥ 0.99; see also Section 4.5), and the same held at the initialization used for the tokenizers we trained (mean resultant length 0.94–0.96, pairwise cosine between the anchors ≥ 0.99).

5.2. The anchor is the leading uncentered direction but not a leading centered one

For every source in every cohort, the anchor coincided with the leading eigenvector of the uncentered second-moment matrix (cosine ≥ 0.997; Figure 2b). Proposition 1 accounts for this almost exactly: the observed sine of the angle (at most 0.074) was a median of 0.92 times the residual bound (4) (at most 0.080), because the cross-covariance between the anchor coordinate

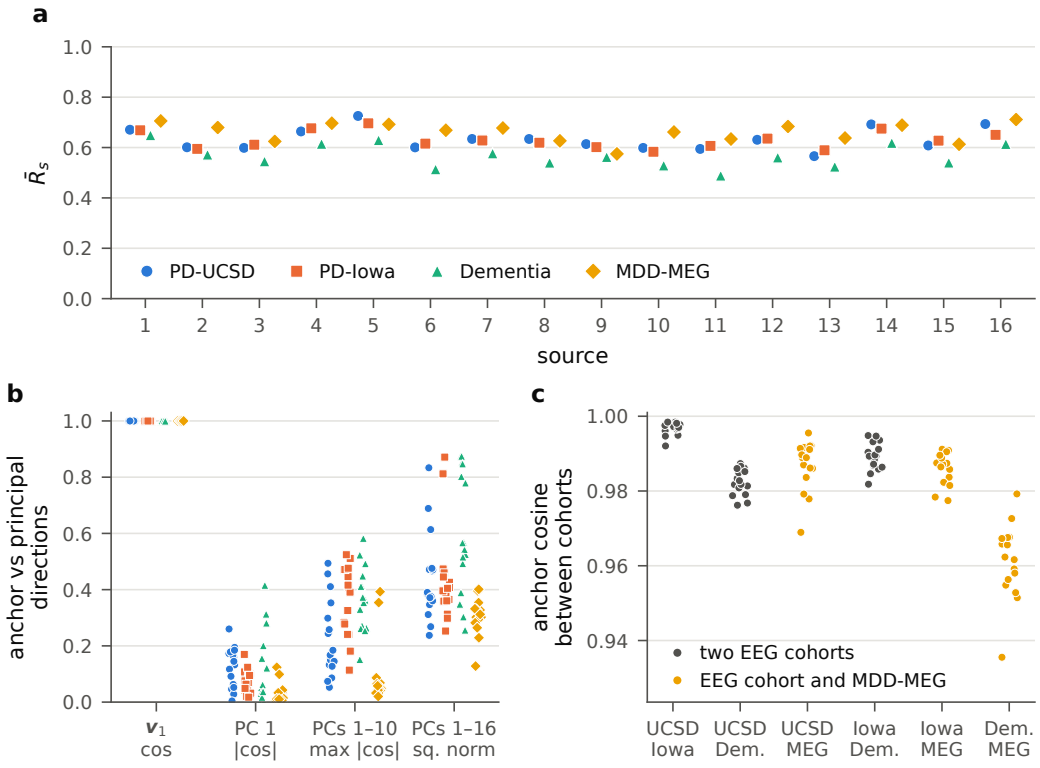


Figure 2: Geometry of the per-source unit vectors (released tokenizer). (a) Mean resultant length $\bar{R}_s$ of each source in each cohort. (b) Anchor versus principal directions, one point per source and cohort: cosine with the leading uncentered direction $\boldsymbol{v}_1$; absolute cosine with the first centered PC; largest absolute cosine with any of the ten leading centered PCs; and squared norm of the anchor's projection onto the span of the 16 leading centered PCs; colors and markers denote cohorts as in (a). (c) Cosine between the anchors of the same source estimated on two different cohorts, for every pair of cohorts (UCSD: PD-UCSD, Iowa: PD-Iowa, Dem.: Dementia, MEG: MDD-MEG); one point per source; grey: two EEG cohorts, amber: an EEG cohort and MDD-MEG.

Table 3: Geometry of the per-source unit vectors (released tokenizer); ranges over the 16 sources. $\bar{R}$: mean resultant length; colat.: mean colatitude; $\cos(\boldsymbol{p}, \boldsymbol{v}_1)$: anchor versus leading uncentered direction; bound: residual bound (4) on $\sin\angle(\boldsymbol{p}, \boldsymbol{v}_1)$; $|\cos(\boldsymbol{p}, \mathrm{PC}_k)|$: largest absolute cosine with the ten leading centered PCs; $\|P_{16}\boldsymbol{p}\|^2$: squared norm of the anchor in the span of the 16 leading centered PCs; var.: share of the centered variance along the anchor (rank among the centered eigenvalues); PR: participation ratio of the centered covariance, $(\operatorname{tr}\Sigma_s)^2/\operatorname{tr}(\Sigma_s^2)$. Upper limits ($\leq$) are rounded up and lower limits ($\geq$) down. Per-source values, with a bootstrap interval for $\bar{R}$, are given in Appendix B.

cohort	$\bar{R}$	**colat.** (°)	$\cos(\boldsymbol{p}, \boldsymbol{v}_1)$	**bound**	$\lvert\cos(\boldsymbol{p}, \mathrm{PC}_k)\rvert$	$\|P_{16}\boldsymbol{p}\|^2$	**var.** (**rank**)	**PR**
PD-UCSD	0.57–0.73	43–55	$\geq$0.9986	$\leq$0.054	$\leq$0.50	0.24–0.83	2.2–3.5% (12–18)	16–21
PD-Iowa	0.58–0.70	45–54	$\geq$0.9994	$\leq$0.038	$\leq$0.53	0.25–0.87	2.4–3.8% (12–17)	16–23
Dementia	0.49–0.65	49–60	$\geq$0.9973	$\leq$0.080	$\leq$0.59	0.26–0.88	2.5–4.9% (7–17)	17–22
MDD-MEG	0.57–0.71	44–55	$\geq$0.9995	$\leq$0.035	$\leq$0.40	0.13–0.40	1.3–1.8% (16–22)	17–20

and the tangent part was small ($\|\boldsymbol{\gamma}_s\|$ between 0.006 and 0.022) compared with the eigengap ($\mathbb{E}[a^2] - \lambda_2$ between 0.18 and 0.49). The anchor coordinate varied little (standard deviation 0.08–0.19), and the bound of (iv), which replaces $\|\boldsymbol{\gamma}_s\|$ by this standard deviation times $\|\Sigma_s\|_{\mathrm{op}}^{1/2}$, already guaranteed a sine of at most 0.25 (cosine ≥ 0.968; the version (iii) of the residual bound was at most 0.080). The classical bound $2\|\Sigma_s\|_{\mathrm{op}}/\bar{R}_s^2$ (0.17–0.68) was 2.5–5.2 times looser than the bound of (iv) and a median of 11 times looser than the residual bound. The anchor did not coincide with any leading centered principal component: its absolute cosine with the first centered PC was at most 0.42 and with any of the ten leading PCs at most 0.59. It was spread instead over several centered PCs, as (v) permits when the eigenvalues of Σ_s near $\mathrm{Var}(a)$ lie closer together than $\|\boldsymbol{\gamma}_s\|$ (per cohort, a median of 12–22 centered eigenvalues lay within $\|\boldsymbol{\gamma}_s\|$ of $\mathrm{Var}(a)$): the best-aligned of the 64 leading PCs, ranked 9th to 33rd, had an absolute cosine with the anchor of 0.40–0.76 (median 0.58). Only 1.3–4.9% of the centered variance lay along the anchor, about as much as the 7th- to 22nd-ranked centered PC carries,

whereas the first centered PC carried 8–15% and the variation was spread over an effective 16–23 dimensions (participation ratio). The anchor is nevertheless not orthogonal to the subspace in which the vectors vary: 0.00–0.76 and 0.13–0.88 of its squared norm lie in the spans of the 10 and 16 leading centered PCs and at least 0.998 in the span of the 64 leading PCs, and the 64 leading tangent axes capture at least 99.7% of the variance of the tangent direction. Apart from the anchor direction, which they exclude by construction, the leading tangent axes span nearly the same subspace as the leading centered PCs (mean squared cosine of the principal angles between the two 16-dimensional spans at least 0.93 for every source and cohort), so the content of Section 5.4 lies essentially in the span of the leading centered PCs (single leading axes can differ from the corresponding PCs where neighboring eigenvalues nearly coincide). The vectors of a source thus occupy a low-dimensional subspace that contains the anchor; within it, the anchor marks where the vectors lie, and the centered PCs describe how they vary around that location.

5.3. Moving a source away from its anchor damages the reconstruction; the exact colatitude carries little information

Table 4 and Figure 3 report the interventions. Moving the vectors of a source to its anchor's equator (zero-anchor; a median geodesic displacement of 35–42°) increased the reconstruction error for every source in every cohort (an interval above zero for all 64 source–cohort pairs, also for the mean absolute error and the correlation), by a median of 3.9–8.1% of the unedited error across cohorts (48% of the damage of replacing the source by another subject's vectors). Halving the anchor coordinate caused 41% of the zero-anchor damage. The effect of the colatitude was asymmetric (Figure 3b):

shifting it by 20° away from the anchor increased the error for 64 of 64 pairs, whereas the same shift towards the anchor slightly decreased it in the EEG cohorts (for 36 of 48 pairs, and increased it for none) and increased it less than the shift away in MDD-MEG (0.6% against 2.4% of the unedited error); the difference between the two directions was positive for 62 of 64 pairs. Moving vectors all the way to the anchor (collapse), which also removes their content, was among the most damaging edits (Table 4). The fluctuations of the anchor coordinate around its mean mattered little: replacing every coordinate by the mean (mean-anchor) or by the coordinate of a randomly permuted vector of the same segment (resample-anchor) caused a median of -2% and 1% of the zero-anchor damage (mean-anchor slightly decreased the error for 36 pairs and increased it for 1; resample-anchor increased it for 18). The reconstruction thus depends on the vectors of each source staying near their anchor—moves away from the anchor are more damaging than moves towards it—but the decoder reads little information from the exact colatitude of a vector.

The damage of zero-anchor is not a consequence of displacement as such. Zero-anchor was 2.4–5.5 times as damaging (cohort medians) as a rotation of the tangent direction that keeps the colatitude and moves each vector by the same angle towards a random direction of the support, and more damaging still than the same rotation towards a random direction of the whole tangent space (zero-anchor was the more damaging edit for all 64 pairs in both comparisons). Zero-anchor moves all vectors of a segment coherently, whereas the matched rotation moves each vector towards its own random direction; coherence alone does not explain the difference, because the coherent shift of the colatitude by 20° towards the anchor did not increase the error in the

EEG cohorts. The decoder is, however, not insensitive to every tangential movement: movement along the leading tangent axes is about as damaging as moving vectors away from the anchor by a similar angle (Section 5.4).

5.4. The content of a source lies along its leading tangent axes

Replacing the tangent direction of a source by that of another subject's vectors, while keeping the colatitude, caused 97% of the damage of replacing the whole source, so the tangent direction carries the content of the source in the segment. This content was concentrated along the leading tangent axes (Figure 3a). Resampling the components on the 1, 4, 16 and 64 leading axes reproduced 16, 49, 96 and 100% of the damage of resampling the whole tangent direction, against 0.3, 1.4 and 9.0% for 1, 4 and 16 random directions of the support (the leading axes were more damaging for all 64 pairs in each of the three comparisons) and 0.2% for 16 random directions of the whole tangent space. The 16 leading axes hold 77–89% of the variance of the tangent direction, and resampling them moved the vectors further than resampling 16 random directions of the support (median displacement 57–62° against 30–32°). The difference is nevertheless not one of displacement: resampling the four leading axes moved the vectors about as far as resampling 16 random support directions (32–35°) and still reproduced 49% of the tangent damage, and at displacements up to about 40°, edits along the leading axes were several times more damaging than edits along random directions of the support, and in the pooled medians at least as damaging as moving the vectors away from the anchor by a similar angle (Figure 3a; in Dementia about equally or less damaging). Mean ablation of the leading axis and of the four leading axes reproduced 7% and 25% of the tangent damage, and collapsing all vectors

Table 4: Change in reconstruction error under each edit, in percent of the unedited mean squared error (median over the 16 sources of the subject-mean change, with a 95% subject-cluster bootstrap interval), and numbers of sources (out of 16) whose bootstrap interval of the subject-mean ΔMSE lies above / below zero, released tokenizer; cells in which any of the three values is below 0.1% in magnitude are given with two decimals. Edits are defined in Table 2; the median unedited errors of the edited segments (on the evaluated sensors) were PD-UCSD 0.39, PD-Iowa 0.40, Dementia 0.31, MDD-MEG 0.33.

edit	**PD-UCSD**	**PD-Iowa**	**Dementia**	**MDD-MEG**
zero-anchor	4.9 [4.4, 5.4] (16/0)	3.9 [3.6, 4.2] (16/0)	6.3 [5.9, 6.8] (16/0)	8.1 [7.7, 8.6] (16/0)
half-anchor	2.1 [1.9, 2.3] (16/0)	1.6 [1.5, 1.7] (16/0)	2.6 [2.4, 2.9] (16/0)	2.8 [2.6, 3.0] (16/0)
mean-anchor	−0.10 [−0.16, −0.04] (1/10)	−0.14 [−0.19, −0.06] (0/11)	−0.3 [−0.4, −0.2] (0/13)	−0.02 [−0.04, 0.00] (0/2)
resample-anchor	0.08 [0.04, 0.12] (7/0)	0.01 [−0.01, 0.04] (1/0)	−0.08 [−0.16, −0.01] (0/2)	0.07 [0.05, 0.09] (10/0)
colatitude −20°	−0.2 [−0.3, −0.1] (0/13)	−0.18 [−0.28, −0.09] (0/11)	−0.7 [−0.8, −0.4] (0/12)	0.6 [0.5, 0.7] (16/0)
colatitude −10°	−0.3 [−0.4, −0.3] (0/16)	−0.3 [−0.4, −0.2] (0/15)	−0.7 [−0.8, −0.6] (0/16)	0.00 [−0.04, 0.02] (3/5)
colatitude +10°	0.8 [0.7, 0.9] (16/0)	0.8 [0.7, 0.8] (16/0)	1.6 [1.4, 1.7] (16/0)	0.5 [0.4, 0.6] (16/0)
colatitude +20°	2.2 [2.0, 2.4] (16/0)	1.8 [1.7, 2.0] (16/0)	3.9 [3.6, 4.2] (16/0)	2.4 [2.3, 2.6] (16/0)
collapse	8.1 [7.1, 9.3] (16/0)	8.2 [7.4, 9.0] (16/0)	12.2 [10.9, 13.4] (16/0)	13.4 [12.6, 14.2] (16/0)
matched rotation (support)	1.4 [1.2, 1.6] (16/0)	1.4 [1.3, 1.6] (15/0)	1.3 [1.1, 1.5] (16/0)	3.0 [2.9, 3.1] (16/0)
matched rotation ($\boldsymbol{p}^{\perp}$)	0.8 [0.7, 0.9] (16/0)	0.7 [0.6, 0.8] (13/2)	0.7 [0.5, 0.8] (16/0)	1.8 [1.7, 2.0] (16/0)
tangent-axis resample (1)	1.8 [1.5, 2.0] (16/0)	1.3 [1.1, 1.4] (16/0)	2.2 [1.9, 2.8] (16/0)	2.5 [2.3, 2.7] (16/0)
tangent-axis resample (4)	5.4 [4.6, 5.9] (16/0)	3.8 [3.4, 4.1] (16/0)	5.7 [5.0, 6.4] (16/0)	9.1 [8.5, 9.6] (16/0)
tangent-axis resample (16)	9.4 [8.3, 10.5] (16/0)	7.9 [7.4, 8.5] (16/0)	12.5 [11.4, 13.6] (16/0)	15.5 [14.6, 16.4] (16/0)
tangent-axis resample (64)	9.9 [8.7, 11.0] (16/0)	8.3 [7.7, 8.9] (16/0)	12.9 [11.9, 14.1] (16/0)	15.9 [15.0, 16.7] (16/0)
random-support resample (16)	0.9 [0.8, 1.0] (16/0)	0.8 [0.7, 0.9] (14/0)	1.3 [1.0, 1.4] (16/0)	1.5 [1.4, 1.6] (16/0)
random resample (16)	0.00 [−0.02, 0.03] (2/0)	0.02 [−0.02, 0.03] (0/2)	0.03 [−0.02, 0.06] (1/2)	0.07 [0.06, 0.10] (11/0)
tangent resample	9.9 [8.7, 11.0] (16/0)	8.3 [7.7, 8.9] (16/0)	12.9 [11.9, 14.1] (16/0)	15.9 [15.0, 16.8] (16/0)
anchor swap (three partners)	7.9 [7.2, 8.3] (16/0)	6.7 [6.3, 7.3] (16/0)	11.5 [10.4, 12.2] (16/0)	13.3 [12.6, 13.9] (16/0)
zero-anchor, keys only	4.4 [4.0, 4.9] (16/0)	3.4 [3.1, 3.7] (16/0)	6.0 [5.5, 6.5] (16/0)	7.9 [7.5, 8.4] (16/0)
zero-anchor, values only	1.1 [1.0, 1.2] (16/0)	0.9 [0.8, 1.0] (16/0)	0.7 [0.6, 0.8] (16/0)	1.1 [1.1, 1.2] (16/0)
anchor swap (first partner)	8.4 [7.6, 9.2] (16/0)	6.5 [6.0, 7.2] (16/0)	11.4 [10.3, 12.6] (16/0)	13.7 [12.5, 14.6] (16/0)
keys only	8.5 [7.7, 9.2] (16/0)	6.5 [6.0, 7.2] (16/0)	11.3 [10.2, 12.7] (16/0)	13.8 [12.6, 14.6] (16/0)
values only	0.08 [0.04, 0.12] (8/1)	0.09 [0.05, 0.11] (9/3)	0.07 [0.04, 0.12] (4/1)	0.08 [0.05, 0.10] (11/0)
source resample	10.3 [9.0, 11.7] (16/0)	8.8 [8.2, 9.5] (16/0)	13.2 [12.2, 14.6] (16/0)	16.2 [15.3, 17.2] (16/0)

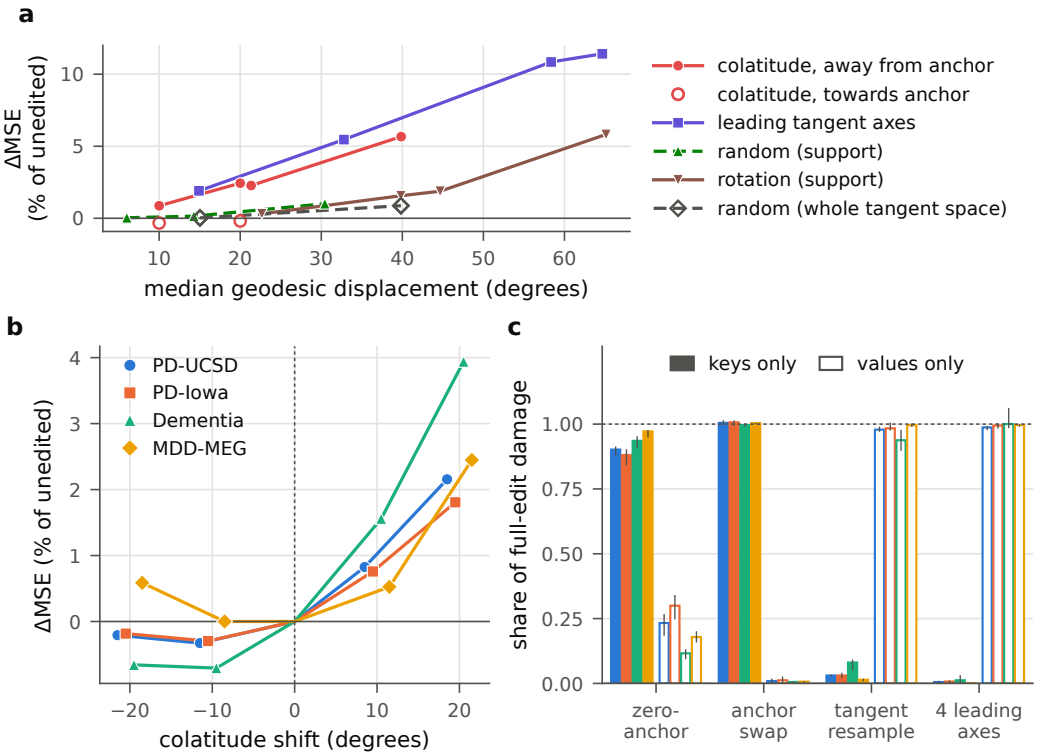


Figure 3: Interventions on one source at a time (released tokenizer). (a) Change in reconstruction error (percent of the unedited error) against the geodesic displacement of the edited vectors, for families of edits (medians over sources and cohorts): colatitude shifts away from the anchor (+10°, half-anchor, +20°, zero-anchor) and towards it (−10°, −20°); resample ablation of the 1, 4, 16 and 64 leading tangent axes and of 1, 4 and 16 random directions of the support; rotations within the support (30°, matched to zero-anchor, 60°, 90°); and the 16-dimensional resample and the matched rotation along random directions of the whole tangent space. (b) Colatitude tuning (median over sources per cohort; points offset horizontally by up to 1.5° for visibility); the point at 0° is the unedited reconstruction. (c) Damage of zero-anchor, the anchor swap, tangent resample and the resample of the four leading tangent axes when the edit enters only the decoder's attention keys (filled bars) or only its values (open bars), relative to the full edit (dashed line); medians over sources with interquartile ranges; colors denote cohorts as in (b).

onto the anchor, which removes all content but also moves the vectors to colatitude zero, caused 76% of the damage of replacing the source.

5.5. Location and identity are read through the decoder's keys, content through its values

The decoder can tell the sources apart only through their quantized vectors (Section 3), which suggests that the anchors act as addresses through which the sensors' attention finds each source. Splitting the edits between the two pathways of the decoder's attention supports this (Figure 3c). Applied only to the input of the keys, with the values computed from the unedited vectors, zero-anchor reproduced a median of 91% of its full damage (cohort medians 88–97%); applied only to the values, it reproduced 19% (the two shares need not add up to 100%; keys more damaging than values for 63 of 64 pairs). Moving a source to its equator reduced the mean attention that the sensors paid to it by a median of 25% (by up to 59%; an interval below zero for 59 of 64 pairs), leaving it 59–86% of the attention that a uniform allocation would give. Rotating all vectors of a source so that its anchor coincided with that of another source, which preserves every angle within the source, caused 81% of the damage of replacing the source by another subject's vectors, and this damage arose almost entirely through the keys (keys only: 100%; values only: 1% of the damage of the swap to the same partner). Content edits showed the opposite pattern: resampling the whole tangent direction reproduced 98% of its damage through the values and 3% through the keys (values more damaging for 64 of 64 pairs), and resampling the four leading tangent axes 99% and 0%. Each source therefore carries a two-part code: where its vectors lie relative to the anchors determines, through the keys, how much the sensors

read from it, and the positions of its vectors along a few leading tangent axes determine, through the values, what they read.

5.6. Tokens: the first level carries the source's identity

Source identity is concentrated in the first token level (Table 5; ranges over cohorts). The level-1 code carried 3.00–3.96 of the 4 bits of source identity (75–99%), and assigning every code to the source that uses it most often recovered the source of 82–99% of the vectors; the deeper levels carried at most 0.15 bits. Identity is nevertheless only part of what the first token encodes: it accounted for 34–45% of the entropy of the level-1 code (8.8–8.9 bits), the rest distinguishing vectors of the same source. Each source used a restricted set of level-1 codewords (median per-source perplexity 29–55 of 512), and these codewords pointed along its anchor (usage-weighted cosine 0.74–0.89 across sources, cohort medians 0.78–0.86); at levels 2–4 every source used nearly the whole codebook (median per-source perplexity 437–475).

The edits changed the codes as the geometry predicts. Moving the vectors of a source to its anchor's equator changed 93% of their level-1 codes, and 99% of the new codes were codes used most often by another source; a colatitude shift of $+20°$ changed 58% of the level-1 codes (78% of them to another source's codes), whereas a shift of $-20°$, towards the anchor, changed 46% almost without leaving the source's vocabulary (median 0%, at most 4%). Content edits also changed level-1 codes mostly within the source's own vocabulary: replacing the tangent direction changed 97% of them, of which 11% went to another source's codes, and among the vectors whose level-1 code was unchanged it changed the level-2 code of 100% (random-support resample of 16 directions: 73%). Moving a vector away from its anchor thus

Table 5: Token structure (released tokenizer). $I(S;C_\ell)$: mutual information between the source index and the level-ℓ code (bits; $H(S) = 4$ bits); accuracy: identification of the source from the level-1 code; perplexity: per-source perplexity of code usage (median over sources; 512 codewords); codeword–anchor cosine: usage-weighted cosine between a source's level-1 codewords and its anchor (median over sources). Lower part: fraction of vectors whose level-1 code changes under an edit and, in parentheses, the fraction of these changes that go to a code used most often by another source (medians over sources).

	PD-UCSD	**PD-Iowa**	**Dementia**	**MDD-MEG**
$I(S;C_1)$ (bits)	3.54	3.51	3.00	3.96
$I(S;C_\ell)$, $\ell = 2,3,4$ (bits)	0.09–0.15	0.05–0.12	0.06–0.13	0.05–0.14
source accuracy from C_1	92%	92%	82%	99%
perplexity, level 1	41	41	55	29
perplexity, levels 2–4	445–454	456–475	442–471	437–449
codeword–anchor cosine	0.82	0.82	0.78	0.86
level-1 codes changed by				
zero-anchor	94% (99%)	93% (99%)	88% (99%)	96% (98%)
colatitude +20°	59% (77%)	58% (78%)	63% (88%)	44% (60%)
colatitude −20°	46% (0%)	45% (1%)	48% (4%)	41% (0%)
mean-anchor	18% (18%)	18% (17%)	24% (34%)	10% (1%)
tangent resample	97% (10%)	97% (12%)	97% (23%)	96% (1%)
tangent-axis resample (16)	96% (8%)	96% (9%)	97% (21%)	96% (1%)
random-support resample (16)	47% (9%)	47% (10%)	50% (21%)	46% (1%)
matched rotation (support)	59% (5%)	59% (5%)	53% (14%)	64% (1%)

moves its first token into the vocabulary of other sources, whereas moving it towards the anchor or changing its tangent direction selects other tokens of the same source.

5.7. The embedded copy and the replicas reproduce the anchors and the identity code; the replicas read location through both pathways

Table 6 and Figure 4 compare the released tokenizer with the copy embedded in the foundation-model checkpoints and with the tokenizers trained from scratch. The embedded tokenizer, whose encoder and decoder equal those of the released tokenizer up to rounding but whose codebooks differ, had the same anchors (per-source cosine ≥ 0.9999 in PD-UCSD and PD-Iowa) and

Table 6: Tokenizers compared: released (all subjects of the four cohorts); embedded, the tokenizer stored in the foundation-model checkpoints (PD-UCSD and PD-Iowa, all subjects; interventions on PD-UCSD only); replicas trained from scratch and the source-embedding variants (held-out subjects of the four cohorts). Ranges over cohorts. MSE: median over all segments of the reconstruction error of a segment on all sensors (the released tokenizer reached 0.37–0.39 on the subjects of the embedded tokenizer and 0.31–0.39 on the held-out subjects of the tokenizers trained from scratch); $\bar{R}$: range over sources and cohorts; anchor dim.: effective number of directions spanned by the 16 anchors (participation ratio of their cosine matrix); zero-anchor: median ΔMSE in percent of the unedited error and number of source–cohort pairs with an interval above zero, out of all pairs evaluated; keys, values: damage of zero-anchor applied to the decoder's keys alone or to its values alone, relative to the damage of the full edit; $I(S; C_1)$: bits.

tokenizer	**MSE**	$\bar{R}$	**anchor dim.**	**zero-anchor**	**keys**	**values**	$I(S; C_1)$
released	0.31–0.39	0.49–0.73	3.9–4.0	3.9–8.1% (64/64)	0.88–0.97	0.12–0.30	3.00–3.96
embedded	0.38–0.40	0.57–0.73	4.0	4.6% (16/16)	0.89	0.23	3.46–3.51
replica 1	0.45–0.58	0.92–0.97	2.0	1.6–2.7% (64/64)	0.79–1.31	1.82–2.48	3.61–3.91
replica 2	0.45–0.55	0.91–0.97	2.0	1.4–3.0% (64/64)	0.51–0.70	1.10–2.40	3.60–3.78
variant 1	0.31–0.43	0.87–0.95	1.0	4.5–10.8% (64/64)	0.00	1.04–1.08	0.06–0.17
variant 2	0.36–0.48	0.91–0.97	1.0	5.1–9.7% (64/64)	0.01–0.02	1.26–1.45	0.34–0.79

the same token structure ($I(S; C_1)$ of 3.46–3.51 bits; zero-anchor changed the level-1 code of 95% of the vectors), and zero-anchor increased its reconstruction error for 16 of the 16 sources in PD-UCSD (median 4.6% of the unedited error; applied to the keys alone, 89% of that damage).

The replicas reconstructed the held-out subjects less accurately than the released tokenizer (median error of a segment 0.45–0.58, against 0.31–0.39 for the released tokenizer on the same subjects), as expected from their much smaller training (Section 4.5). Their sources nevertheless developed the structure described above, starting from an initialization at which all sources coincided (Section 5.1). In every replica and cohort, the vectors of each source were concentrated in a narrow cap around their anchor (in replica 2 the anchors of two sources nearly coincided; see below), and the anchors estimated on different cohorts agreed (per-source cosine ≥ 0.997; all

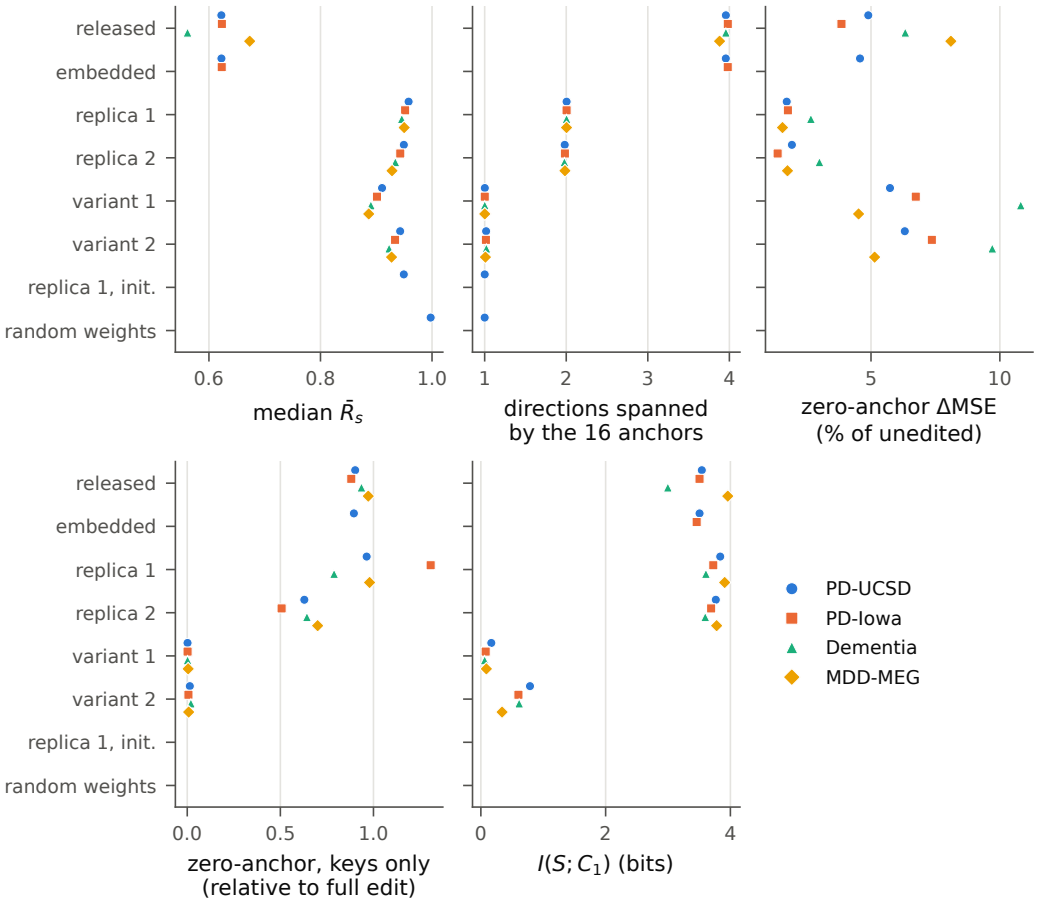


Figure 4: Key quantities for every tokenizer examined (rows) and cohort (colors and markers): median mean resultant length of the sources; number of directions spanned by the 16 anchors (participation ratio of their cosine matrix); median damage of zero-anchor in percent of the unedited error; median damage of zero-anchor applied to the decoder's keys alone, relative to the full edit; and $I(S; C_1)$. The tokenizers trained from scratch were evaluated on held-out subjects; the untrained controls (initialization of replica 1; released architecture with random weights) are shown for the geometry only (embedded: PD-UCSD and PD-Iowa, interventions on PD-UCSD only; untrained controls: PD-UCSD).

four cohorts contributed training data, so this agreement is expected). The anchor was the leading uncentered direction (cosine ≥ 0.999) but carried only 0.7–7.1% of the centered variance, about as much as the 7th- to 13th-ranked centered PC (absolute cosine with the first centered PC at most 0.09). Three features of the geometry differed from the released tokenizer. The caps were narrower ($\bar{R}_s$ between 0.91 and 0.97; mean colatitude 12–24°). The anchors were arranged on a circle rather than in four dimensions: in every replica and cohort, two directions held at least 99.7% of the squared norm of the 16 anchors. In replica 1, consecutive anchors along the circle were 15–27° apart, close to the 22.5° of an even spacing; in replica 2, the anchors of two sources nearly coincided (1° apart in every cohort), so that its 16 sources occupied 15 positions on the circle (largest gap 31°). And the anchor was closer to a single centered PC, ranked 7th to 12th (median absolute cosine 0.95), consistent with the argument after Proposition 1: fewer centered eigenvalues lay near $\mathrm{Var}(a)$ (per cohort, a median of 1–3 within $\|\boldsymbol{\gamma}_s\|$ of it, against 12–22 in the released tokenizer), whereas $\|\boldsymbol{\gamma}_s\|$ relative to $\mathrm{Var}(a)$ was similar (median 0.47 against 0.63).

The causal roles replicated in part. Zero-anchor increased the reconstruction error for 128 of the 128 source–cohort pairs of the two replicas (median 1.4–3.0% of the unedited error across cohorts and replicas), rotating a source onto another source's anchor acted through the keys (100% of the damage) rather than the values (3%), and the content again lay along the leading tangent axes: resampling 16 of them reproduced 100% of the damage of tangent resample, resampling 16 random directions of the same support 9.5%, and content edits acted through the values (96% of the damage of tangent

resample) rather than the keys (2%). The location code, however, differed. Applied to the keys alone, zero-anchor caused a median of 69% of the damage of the full edit, and the attention paid to the source fell by a median of 36%; applied to the values alone, it was even more damaging than the full edit (196%), so that the keys were the more damaging pathway for only 6 of the 128 pairs, and the decoders of these tokenizers read the location of a source through both pathways. In their narrow caps zero-anchor moves the vectors further (69–75° against 35–42°), and the matched rotation, mostly the 180° fallback here, was the more damaging edit for every pair, so it is no displacement control for these tokenizers. The exact colatitude also carried information: replacing the anchor coordinate by its mean or by that of a permuted vector increased the error for 103 and 126 pairs (9% and 17% of the zero-anchor damage), and shifts towards the anchor were damaging: a shift of 10° increased the error for 117 pairs, although a median of only 9% of the vectors of a source lie within 10° of the anchor, and a shift of 20° was more damaging than the same shift away, as it moves the vectors within 20° of the anchor (a median of 69%) onto the anchor itself. The first RVQ level again carried most of the information about source identity: $I(S; C_1)$ was 3.60–3.91 bits (41–45% of the entropy of the level-1 code), and zero-anchor changed the level-1 code of 99.7% of the vectors, 99% of them to a code used most often by another source.

5.8. With a source embedding in the decoder, the anchors do not separate

The source-embedding variants test whether the organization described above follows from the decoder's lack of a source index. Their decoders receive a learned embedding of each source, and their training, from the same

initial weights and batches as the replicas, did not separate the sources by location: in both variants the vectors of each source were again concentrated ($\bar{R}_s$ between 0.87 and 0.97), but around nearly one common direction. In every cohort each anchor lay within 9° of the mean direction of the 16 anchors, less than the mean colatitude of any source (13° to 29°), so that the caps of all sources overlapped; any two anchors lay within 17° of each other (pairwise cosine ≥ 0.95), whereas in the replicas they lay up to 177° apart. Accordingly, rotating a source onto another source's anchor moved its vectors by only 2–7° (81–89° in the replicas) and caused 0.3% of the damage of replacing the source by another subject's vectors (82% in the replicas), and the first token level encoded little of the source identity ($I(S; C_1)$ of 0.06–0.79 bits, 1–9% of the entropy of the level-1 code, against 3.60–3.91 of the 4 bits in the replicas). Variant 2, whose anchors were more dispersed, also encoded more identity (any two anchors within 17° and 0.34–0.79 bits, against 5° and 0.06–0.17 bits in variant 1). Moving the vectors of a source away from the common anchor still damaged the reconstruction (zero-anchor: 128 of 128 pairs, median 4.5–10.8% of the unedited error), but almost only through the values: applied to the keys alone, zero-anchor caused a median of 0.6% of its damage and applied to the values alone 117%, and the attention paid to the edited source fell by a median of only 2%. Content edits likewise acted through the values (tangent resample: 100% of its damage through the values, 0.3% through the keys), and the content again lay along the leading tangent axes (resampling 16 of them: 100% of the damage of tangent resample; 16 random support directions: 8.5%). In these tokenizers the keys thus hardly respond to the quantized vectors, and the sensors find each source through its

embedding. In every cohort, each variant reconstructed the held-out subjects better than the replica of the same seed (median error of a segment 0.31–0.48 against 0.45–0.58; released tokenizer on the same subjects 0.31–0.39) and fitted its training batches better (correlation between reconstructed and target waveforms at the end of training 0.71–0.74 against 0.68).

5.9. Diagnostic information is present in the sources but, within cohorts, not concentrated in the anchors

Table 7 and Figure 5 summarize the subject-level read-outs. In the three contrasts in which band power distinguished patients from controls—Alzheimer's disease (AD; AUC 0.83), frontotemporal dementia (FTD; 0.83) and Parkinson's disease at the Iowa site (0.73)—the full source representation and the per-source means of the quantized vectors did so as well (full 0.83, 0.76 and 0.73; quantized 0.82, 0.75 and 0.71; all significant after correction). In AD and PD-Iowa their AUCs were close to those of band power (paired difference of the full representation 0.00 [−0.04, 0.04] and 0.00 [−0.06, 0.05]), whereas in FTD their point estimates were lower (full −0.07 [−0.16, 0.001]; quantized −0.08 [−0.17, −0.002]; descriptive intervals with the out-of-fold scores held fixed, Section 4.6), a difference these data do not establish. The tokenizer thus retains much of the diagnostic information that band power captures, with a possible loss in FTD. In PD-UCSD (31 subjects) and MDD-MEG (49 subjects), no representation separated the groups, band power included. Age and sex alone separated the groups at most weakly (AUC 0.57 in AD, 0.54 in FTD and 0.58 in PD-Iowa, $p = 0.046$ uncorrected; 0.64 in MDD-MEG, $p = 0.040$ uncorrected), and regressing them out of the features inside every training fold left the full representation and band power above chance (full:

0.79, 0.78 and 0.73; band power: 0.81, 0.83 and 0.72). In PD-UCSD, where age and sex are matched between the groups, the classifier on age and sex alone was below chance (0.31), as can occur in cross-validation when covariates are more closely matched between the groups than random labels would make them (Parker et al., 2007; Airola et al., 2011); its permutation statistic lay in the lower tail of the null distribution of the same pipeline (null mean 0.48, $p = 0.878$), so the test was not misled.

The anchor coordinates carried less of this information. Anchor-16 was above chance after Holm correction across the five contrasts in two of them (AD: 0.72 [0.61, 0.82], $p = 0.036$; PD-Iowa: 0.67 [0.59, 0.75], $p = 0.015$) but not in FTD (0.65 [0.52, 0.77], $p = 0.084$). In all three its AUC was below that of the full representation (paired difference full minus anchor-16: 0.12 [0.05, 0.19] in AD, 0.06 [−0.01, 0.12] in PD-Iowa and 0.11 [−0.001, 0.22] in FTD), and 16 random directions of the same per-source representation, evaluated identically, reached at least the same AUC in 84%, 27% and 59% of 100 draws (8% in PD-UCSD and 86% in MDD-MEG; Monte Carlo $p \geq 0.089$ in every contrast). Within cohorts, the anchors are therefore not a privileged summary of the diagnostic information. Individually, the anchor coordinate of source 4 separated patients from controls after family-wise correction across the 16 sources of each contrast in both AD (AUC 0.76, $p < 0.001$) and FTD (0.74, $p = 0.016$), which share their control group; in total, 5 sources did so in AD, 1 in FTD and 1 in PD-Iowa (source 1, 0.64, $p = 0.034$), and none in PD-UCSD or MDD-MEG. Selecting the best source inside every training fold, which accounts for the selection, gave AUCs of 0.67 in AD and 0.70 in FTD (Table 7).

Table 7: Subject-level diagnostic read-out (released tokenizer): AUC (mean over 20 repetitions of stratified five-fold cross-validation) with a 95% stratified bootstrap interval over subjects in which the out-of-fold scores are held fixed (it does not include the variability of model fitting; inference rests on the permutation tests). Markers: * permutation $p < 0.05$ (anchor-16: Holm-corrected across contrasts; secondary representations: Benjamini–Hochberg-corrected within contrast; best single anchor, adjusted representations and age and sex: uncorrected). The permutation tests use the pipeline with the inverse regularization strength fixed at 1 and two repetitions (Section 4.6), whose statistic differs from the AUC shown by at most 0.12, so a marked AUC can have an interval that includes 0.5. Best single anchor: the anchor coordinate whose AUC on the training subjects is farthest from 0.5, selected inside every training fold and evaluated out of fold (per-source results are given in the text); adjusted: age and sex regressed out within training folds. Last row: AUC range (2.5–97.5% quantiles) of 100 random 16-direction representations evaluated like anchor-16.

	PD-UCSD **PD vs HC** ($n = 31$)	**PD-Iowa** **PD vs HC** ($n = 148$)	**Dementia** **AD vs CN** ($n = 65$)	**Dementia** **FTD vs CN** ($n = 52$)	**MDD-MEG** **MDD vs HV** ($n = 49$)
anchor-16	0.62 [0.48, 0.76]	0.67* [0.59, 0.75]	0.72* [0.61, 0.82]	0.65 [0.52, 0.77]	0.42 [0.31, 0.55]
best single anchor	0.44 [0.31, 0.57]	0.59 [0.51, 0.67]	0.67* [0.57, 0.78]	0.70* [0.56, 0.83]	0.47 [0.36, 0.59]
quantized anchor-16	0.53 [0.38, 0.67]	0.70* [0.61, 0.78]	0.75* [0.64, 0.85]	0.63* [0.49, 0.75]	0.49 [0.36, 0.62]
concentration-16	0.59 [0.44, 0.73]	0.69* [0.61, 0.77]	0.68* [0.57, 0.79]	0.65* [0.52, 0.77]	0.42 [0.31, 0.55]
norm-16	0.63 [0.50, 0.76]	0.64* [0.56, 0.72]	0.79* [0.68, 0.89]	0.71* [0.58, 0.83]	0.42 [0.30, 0.55]
pooled	0.56 [0.40, 0.73]	0.68* [0.61, 0.76]	0.83* [0.73, 0.91]	0.62* [0.50, 0.74]	0.50 [0.38, 0.62]
tangent	0.49 [0.33, 0.65]	0.73* [0.66, 0.80]	0.83* [0.72, 0.91]	0.77* [0.65, 0.88]	0.53 [0.39, 0.66]
full	0.52 [0.35, 0.68]	0.73* [0.66, 0.80]	0.83* [0.73, 0.91]	0.76* [0.64, 0.87]	0.52 [0.39, 0.65]
quantized	0.48 [0.32, 0.65]	0.71* [0.63, 0.79]	0.82* [0.72, 0.91]	0.75* [0.63, 0.86]	0.54 [0.40, 0.67]
band power	0.52 [0.39, 0.66]	0.73* [0.65, 0.80]	0.83* [0.74, 0.91]	0.83* [0.73, 0.93]	0.49 [0.35, 0.62]
anchor-16, adjusted	0.63 [0.50, 0.77]	0.68* [0.60, 0.76]	0.70* [0.59, 0.81]	0.62* [0.50, 0.73]	0.50 [0.38, 0.62]
full, adjusted	0.52 [0.35, 0.69]	0.73* [0.64, 0.80]	0.79* [0.68, 0.89]	0.78* [0.66, 0.89]	0.50 [0.38, 0.62]
band power, adjusted	0.49 [0.37, 0.62]	0.72* [0.63, 0.79]	0.81* [0.71, 0.90]	0.83* [0.71, 0.92]	0.53 [0.40, 0.66]
age and sex	0.31 [0.26, 0.35]	0.58* [0.51, 0.65]	0.57 [0.48, 0.67]	0.54 [0.43, 0.66]	0.64* [0.52, 0.76]
random 16 directions	0.38–0.66	0.56–0.74	0.66–0.86	0.53–0.81	0.37–0.62

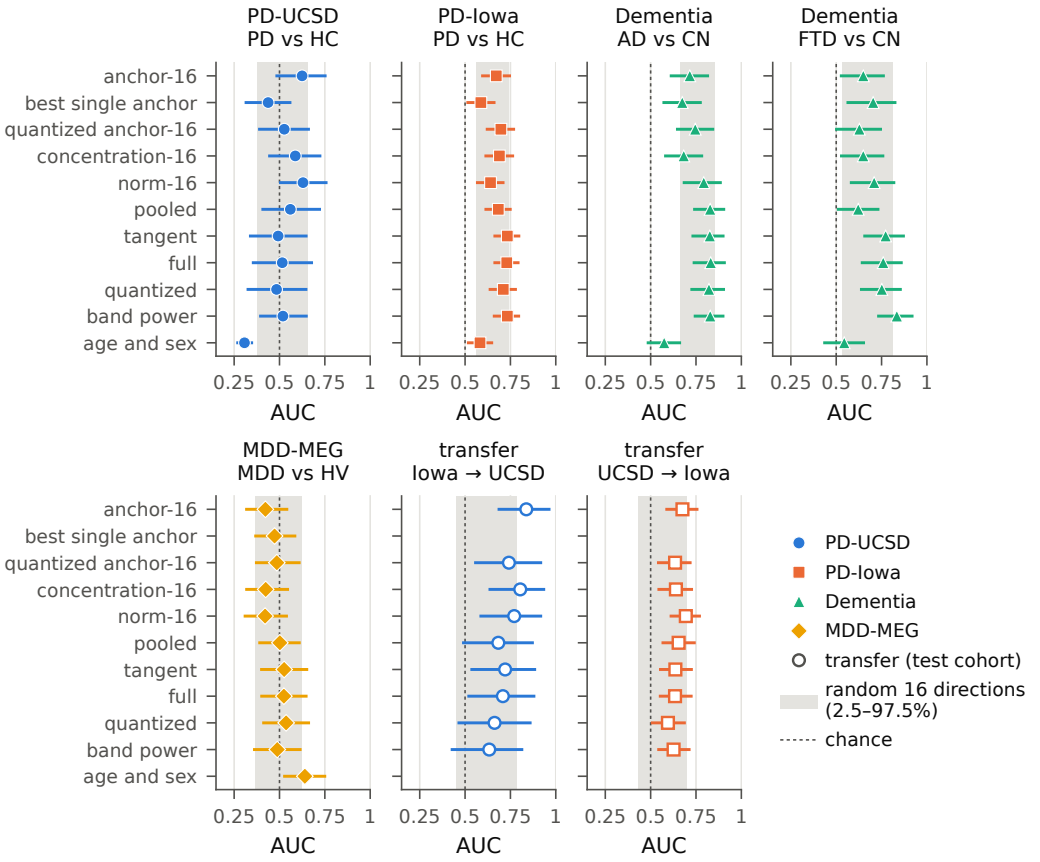


Figure 5: Subject-level diagnostic read-out (released tokenizer): AUC with 95% bootstrap interval (within-cohort panels: mean over 20 repetitions of stratified five-fold cross-validation, out-of-fold scores held fixed in the bootstrap; transfer panels: AUC of a single classifier on the test cohort, bootstrap over test subjects); colors denote cohorts. The last two panels (bottom row; open markers, colored by the cohort tested) show transfer between the Parkinson's cohorts: a classifier trained on all subjects of one cohort and tested on the other, with the training cohort's feature scaling; band power is restricted to the shared sensors; rows without a value (best single anchor, age and sex) were not evaluated for transfer. The grey band is the 2.5–97.5% range of the AUCs of random 16-direction representations evaluated like anchor-16 (100 draws within cohorts, 1,000 in the transfer panels); the dashed line marks chance.

Table 8: Transfer between the two Parkinson's cohorts (released tokenizer): a classifier trained on all subjects of one cohort and applied to the other. AUC with 95% bootstrap interval over the test subjects; p: permutation of the test labels, Holm-corrected across the two directions for anchor-16 and Benjamini–Hochberg-corrected within a direction for the other representations; rand.: number of the 1,000 random 16-direction representations with at least the same AUC. Band power is restricted to the 29 sensors shared by the two montages.

	PD-Iowa → PD-UCSD			PD-UCSD → PD-Iowa		
representation	AUC	p	rand.	AUC	p	rand.
features standardized with the training cohort's statistics						
anchor-16	0.84 [0.68, 0.97]	<0.001	2	0.68 [0.58, 0.76]	<0.001	62
quantized anchor-16	0.74 [0.55, 0.93]	0.030	88	0.63 [0.53, 0.73]	0.006	202
concentration-16	0.80 [0.63, 0.94]	0.017	13	0.64 [0.54, 0.73]	0.006	188
norm-16	0.77 [0.58, 0.93]	0.020	48	0.69 [0.60, 0.78]	<0.001	28
pooled	0.68 [0.48, 0.88]	0.057	252	0.65 [0.56, 0.75]	0.006	128
tangent	0.72 [0.53, 0.89]	0.043	131	0.64 [0.55, 0.73]	0.006	197
full	0.71 [0.51, 0.89]	0.043	158	0.63 [0.55, 0.73]	0.006	203
quantized	0.66 [0.46, 0.87]	0.077	342	0.60 [0.50, 0.69]	0.030	417
band power	0.63 [0.42, 0.82]	0.110	492	0.63 [0.54, 0.72]	0.007	244
features standardized within each cohort						
anchor-16	0.81 [0.64, 0.95]	0.002	13	0.71 [0.62, 0.80]	<0.001	16
quantized anchor-16	0.74 [0.55, 0.91]	0.035	82	0.70 [0.60, 0.79]	<0.001	28
concentration-16	0.77 [0.58, 0.92]	0.019	40	0.64 [0.54, 0.74]	0.005	202
norm-16	0.78 [0.58, 0.93]	0.019	30	0.69 [0.60, 0.78]	<0.001	39
pooled	0.58 [0.38, 0.80]	0.229	702	0.66 [0.57, 0.76]	0.003	115
tangent	0.70 [0.50, 0.89]	0.053	185	0.64 [0.55, 0.73]	0.005	229
full	0.71 [0.52, 0.89]	0.049	141	0.64 [0.55, 0.74]	0.005	192
quantized	0.64 [0.43, 0.85]	0.112	408	0.60 [0.51, 0.70]	0.022	393
band power	0.66 [0.46, 0.85]	0.099	340	0.62 [0.53, 0.72]	0.008	291

Across sites (Table 8), the representations transferred above chance in most cases, and the anchor coordinates stood out against random directions in one of the two directions with the training cohort's feature scaling and in both with features standardized within each cohort. A classifier trained on all PD-Iowa subjects and applied to PD-UCSD, recorded at another site with another montage, reached an AUC of 0.84 [0.68, 0.97] with the anchor coordinates ($p < 0.001$), higher than all but 2 of 1,000 representations by 16 random directions (Monte Carlo $p = 0.006$, Holm-corrected across the two directions; 2.5–97.5% range 0.45–0.79); trained on the 31 PD-UCSD subjects

and applied to PD-Iowa, the anchor coordinates reached 0.68 [0.58, 0.76] ($p < 0.001$), in the upper tail of the random directions (62 of 1,000 at least as high; Monte Carlo $p = 0.063$). With the features standardized within each cohort, the anchor coordinates reached 0.81 and 0.71, above the random directions in both directions (Monte Carlo $p = 0.028$ and $p = 0.028$, Holm-corrected). With Holm correction across both scalings and both directions (four tests), the Monte Carlo p-values are $p = 0.012$ and $p = 0.063$ with the training cohort's scaling and $p = 0.042$ and $p = 0.042$ with the features standardized within each cohort, so each conclusion is unchanged. With one exception (below), the anchor coordinates were nevertheless not shown to transfer better than the other representations, all of which also transferred from PD-UCSD to PD-Iowa (largest corrected $p = 0.030$ with the training cohort's scaling and $p = 0.022$ with the features standardized within each cohort) and some of which transferred from PD-Iowa to PD-UCSD. With the training cohort's feature scaling no paired difference between the anchor coordinates and another representation was significant after correction, the largest being those to band power (0.20, unadjusted interval [0.01, 0.41], Holm-corrected $p = 0.228$) and to the quantized representation (0.18, [0.05, 0.32], $p = 0.056$) from PD-Iowa to PD-UCSD; with the features standardized within each cohort, the only significant one was that to concentration-16 from PD-UCSD to PD-Iowa (0.07, [0.03, 0.12], Holm-corrected $p = 0.008$). The other per-source scalar summaries, concentration-16 and norm-16, transferred in both directions and lay in the upper tail of the random directions from PD-Iowa to PD-UCSD, norm-16 also from PD-UCSD to PD-Iowa (Table 8).

6. Discussion

6.1. Anchors as addresses, tangent axes as content

The latent sources of BrainOmni's tokenizer are not interchangeable channels with arbitrary content. Each source occupies a cap around its own learned anchor, which is nearly the same for every montage, site, population and modality we examined; the caps overlap (the smallest angle between two anchors is below the mean colatitude), and the vectors of a source carry a two-part code. The location of the vectors relative to the anchors determines, through the decoder's attention keys, how much the sensors read from a source: vectors moved towards the equator receive less attention, and rotating them onto another source's anchor damages the reconstruction almost entirely through the keys, consistent with their being read as if they came from another source. The content lies in the tangent direction, concentrated along a few leading tangent axes, and reaches the reconstruction through the attention values; the exact colatitude carries little of it. The anchor direction is thus an address rather than a carrier of content: it resembles the constant offsets and outlier directions that act as biases in language and vision transformers (Biś et al., 2021; Timkey and van Schijndel, 2021; Sun et al., 2024; Darcet et al., 2024), except that its role is to keep the sources apart. These addresses occupy few of the 256 dimensions: the 16 anchors lie close to a four-dimensional subspace in the released tokenizer and close to a circle in the replicas we trained (Section 5.7).

The architecture suggests why such a code arises. Every source is an attention-weighted average of the same per-sensor features passed through shared value and output projections, the quantizer discards its norm and uses

codebooks shared by all sources, and the decoder receives neither a source index nor a source embedding. The quantized vectors themselves are therefore the only channel through which the decoder can learn which source a vector belongs to. With the released initialization all sources point in the same direction; the released training separated them (at our much smaller training scale the sources did not separate in a pilot run with this initialization, which we therefore modified; Section 4.5), and the first RVQ level then carries most of the information about source identity, although identity is less than half of what a level-1 token encodes. The source-embedding variants support this account (Section 5.8): when the decoder could identify the sources through a learned embedding, training from the replicas' initial weights and batches left all sources around nearly one common direction (so the modified initialization does not by itself separate them), the first token level encoded little source identity, and the keys hardly responded to the quantized vectors. Separated anchors, the reading of identity through the keys and the identity code of the first token level thus appear as three consequences of one constraint, that the quantized vectors are the decoder's only access to source identity; the evidence comes from two variants, each paired with a replica of the same seed, trained at a small scale. For BrainOmni's foundation model this identity information is redundant: the model adds the encoder's learned query vector of each source to that source's tokens. Subtracting the learned anchors before quantization or giving each source its own first-level codebook are natural ways to reclaim the capacity; whether they improve downstream performance is a question for future work.

6.2. Implications for interpreting representations

Table 9 sets what three standard analyses suggest about the anchors of the released tokenizer against what the analyses used here show. Four methodological points generalize beyond this model. First, for a representation concentrated around a mean direction, the leading eigenvector of the uncentered second moment lies close to that direction whenever the cross-covariance between the coordinate along it and the orthogonal part is small compared with the gap between the mean squared coordinate and the second eigenvalue (Proposition 1); a coordinate that varies little guarantees this. The centered principal components instead describe the variation around the mean direction (Mu and Viswanath, 2018; Honeine, 2014). Statements that a mean direction "is the top principal component", and control directions chosen by principal-component rank, must therefore name the second moment they refer to. Second, edits of normalized representations should preserve the normalization: some edits that move vectors off the sphere, such as scaling or averaging the coordinate along a direction, are partly undone when the model renormalizes them, so that the effect attributed to the edit is smaller than, and not proportional to, the edit itself (halving the anchor component off the sphere would have achieved 70–89% of the intended change in the released tokenizer and only 10–36% in the narrow caps of the replicas). Third, projection (zero) ablation of a mean direction removes a location as well as any information carried along it. Mean and resample ablation separate the two (Chan et al., 2022; Heimersheim and Nanda, 2024; Zhang and Nanda, 2024; Li and Janson, 2024), and on the sphere, controls must also be matched in displacement and in support, because the same displacement is nearly

harmless along some directions and damaging along others. In our data, zero ablation alone would attribute about half of the effect of replacing a source (48%) to a single direction; mean and resample ablation show that this direction carries little information, displacement- and support-matched controls identify the directions that do, and the decomposition of the decoder's attention into keys and values (Elhage et al., 2021) shows why the location matters. Fourth, diagnostic read-outs from foundation-model representations of small cohorts must be evaluated at the level of subjects with calibrated null distributions: cross-validated estimates of the AUC are biased downwards and can fall below 0.5 when there is no signal, most strongly for pooled leave-one-out estimates (Parker et al., 2007; Airola et al., 2011), which is why inference should rest on permutation nulls that re-run the same pipeline; segment-level splits measure subject identity (Saeb et al., 2017; Lin et al., 2026); and a candidate summary should be compared with random summaries of the same dimension.

6.3. Diagnostic content

The read-outs separate two questions that are easily conflated. Whether the tokenizer preserves diagnostic information is answered by the full source representation and its quantized counterpart, which separated the groups in every contrast in which band power was informative, with AUCs close to those of band power in AD and PD-Iowa and somewhat lower in FTD; for these resting-state contrasts the tokenizer retains much of the diagnostic information, although a loss in FTD cannot be excluded. Whether that information is concentrated in the anchors is answered by comparing the 16 anchor coordinates with random 16-dimensional projections of the same

Table 9: Three standard analyses of the anchors of the released tokenizer, what they suggest, and what the analyses used here show.

question	**standard analysis and what it suggests**	**analysis used here and what it shows**
Is the anchor a principal direction?	Centered PCA: the anchor carries 1.3–4.9% of the variance of a source (ranked 7th to 22nd), a minor direction.	Uncentered second moment and Proposition 1: the anchor is the leading direction of every source (cosine ≥ 0.997) and fixes where its vectors lie (Section 5.2).
Is information carried along the anchor?	Projection (zero) ablation of the anchor: 48% of the damage of replacing the source by another subject's vectors, as if half of the information of a source lay along one direction.	Mean and resample ablation of the anchor coordinate: -2% and 1% of the zero-anchor damage; zero-anchor is 2.4–5.5 times as damaging as rotations matched in displacement; the damage is a change of location, read through the decoder's keys (91%; Sections 5.3 and 5.5).
Are the anchor coordinates a privileged diagnostic summary within a cohort?	Linear probe tested against chance: above chance in two of five contrasts (AD and PD-Iowa), as if the anchors summarized the diagnostic information.	Probe tested against random 16-direction projections of the same representation: within cohorts, these reach at least the same AUC in 84% and 27% of draws in AD and PD-Iowa (Monte Carlo $p \geq 0.089$ in every contrast); in transfer between the Parkinson's cohorts, the anchor coordinates exceed them in at least one direction (Section 5.9).

representation, and the anchor coordinates did not do significantly better. This is consistent with their role as addresses: a subject-level shift of the colatitude of a source changes how strongly the decoder attends to it but carries little content. The source whose anchor coordinate separated both dementia groups from controls shows that such shifts can nevertheless covary with disease. Across sites the picture changed: with the training cohort's feature scaling, the anchor coordinates trained on PD-Iowa transferred to PD-UCSD better than random projections of the same representation, whereas

in the other direction they lay in the upper tail of the random projections, and with the features standardized within each cohort they exceeded the random projections in both directions; the other per-source scalar summaries (concentration and norm) also transferred, and with the training cohort's feature scaling none of their differences from the anchor coordinates, nor those of the full representation or band power, was significant. A plausible reading is that how strongly each source is engaged—how close its vectors lie to the anchor, how concentrated and how large they are—is less affected by site and montage than the content of the sources or the band power of individual sensors. Because one side of each comparison is a cohort of 31 subjects, this observation needs confirmation in further cohorts. Neither the source representation nor band power separated the groups in PD-UCSD, the smallest cohort, or in MDD-MEG, in which age and sex alone were weakly predictive; we draw no conclusion about these contrasts beyond the absence of a detectable effect.

6.4. Limitations

The analysis concerns one tokenizer architecture. We examined the released checkpoint, the differently quantized copy inside the foundation-model checkpoints and four tokenizers trained from scratch, but the latter were trained with a much smaller corpus, batch and budget than the released tokenizer and with a modified recipe (Section 4.5), and the replicas reached a higher reconstruction error; other normalized-RVQ tokenizers may organize their latent space differently. Our causal criterion is the tokenizer's own reconstruction; we did not test how the anchors affect the BrainOmni foundation model that consumes the tokens or its downstream accuracy. Edits

were applied to one source at a time and to the quantizer input; interactions between sources and edits of the encoder were not examined. One of the cohorts (PD-UCSD) appears in the configuration of the released pretraining code and may have been part of the tokenizer's self-supervised training data; the geometric and causal findings were the same on the three other cohorts, and the tokenizers we trained, analyzed on held-out subjects only, are compared in Sections 5.7 and 5.8. For MEG we analyzed the first 180 s of each recording. The diagnostic analyses involve modest cohorts with one or two diagnostic contrasts each, characterize representations rather than clinical utility, and were not validated in independent clinical samples beyond the transfer between the two Parkinson's cohorts. Finally, we describe where the anchors lie and how the decoder uses them, but not how the encoder's attention produces them, which is a natural next step for a circuit-level account.

7. Conclusion

The 16 latent sources of BrainOmni's tokenizer form a structured code. Each source lives in a cap around a learned anchor that is largely fixed by the tokenizer and shared across cohorts and modalities; the anchor is the source's leading uncentered principal direction but carries only a few percent of its centered variance, no more than a typical one of its roughly 20 effective dimensions; the decoder's attention keys use the location of the vectors relative to the anchors to decide how much to read from a source, and its values read the source's content from a few leading tangent axes; and the first token level carries most of the information about which source a vector

belongs to. Two replicas trained from scratch reproduce the anchors, the reading of identity through the keys, the content along leading tangent axes and the identity code, although they read the location of a source through both attention pathways; two variants whose decoder receives an embedding of each source do not separate the anchors and read almost nothing through the keys, which supports the decoder's lack of a source index as the origin of the code. In subject-level read-outs, the source representation and its tokens carry diagnostic information wherever band power does, and this information is not concentrated in the anchors within a cohort, although the anchor coordinates transferred between the two Parkinson's cohorts better than random directions, in one direction with the training cohort's feature scaling and in both with features standardized within each cohort; with 31 subjects in one of the two cohorts, this needs confirmation in further cohorts. These conclusions rest on interventions that preserve the geometry imposed by the quantizer and on controls matched in displacement, dimension and support, and the same methodology can be applied to other models that normalize their latent vectors before quantizing or attending to them.

Ethics statement

This work uses only publicly available, de-identified datasets distributed under CC0 licenses through OpenNeuro, collected under the ethical approvals of the original studies. No new data were collected. The diagnostic analyses characterize tokenizer representations and are not intended or validated for clinical use.

CRediT authorship contribution statement

Ihor Kendiukhov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing — original draft, Writing — review & editing.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work the author used Claude Opus 4.7 and Claude Opus 5.5 (Anthropic) to assist with code development, running the analyses, and drafting and editing the text. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Data availability

All datasets are public on OpenNeuro: ds002778, ds004584, ds004504 and ds003568. Code for preprocessing, activation export, geometry, interventions,

token analysis, diagnostic read-out, tokenizer training and figure generation and the subject and segment lists are available at `https://github.com/Biodyn-AI/neurofm-mechinterp` (release v1.0.0); the intermediate results needed to regenerate every number, table and figure, including the weights of the tokenizers we trained, are attached to that release.

Appendix A. Proof of Proposition 1

Write $\boldsymbol{p} = \boldsymbol{p}_s$, $M = M_s$, $\Sigma = \Sigma_s$, $\boldsymbol{\mu} = \boldsymbol{\mu}_s$, $\boldsymbol{\gamma} = \boldsymbol{\gamma}_s$ and $\bar{R} = \bar{R}_s$, and decompose $\boldsymbol{u} = a\boldsymbol{p} + \boldsymbol{t}$ with $\boldsymbol{t} \perp \boldsymbol{p}$.

(i) Because $\mathbb{E}[a] = \langle \boldsymbol{\mu}, \boldsymbol{p} \rangle = \bar{R}$ and $\boldsymbol{\mu} = \bar{R}\boldsymbol{p}$, the tangent part has mean $\mathbb{E}[\boldsymbol{t}] = \boldsymbol{\mu} - \mathbb{E}[a]\boldsymbol{p} = \boldsymbol{0}$. Hence $\boldsymbol{\gamma} = \mathrm{Cov}(a, \boldsymbol{t}) = \mathbb{E}[a\boldsymbol{t}]$, which is orthogonal to $\boldsymbol{p}$ because every $\boldsymbol{t}$ is. Then $M\boldsymbol{p} = \mathbb{E}[\boldsymbol{u}\langle \boldsymbol{u}, \boldsymbol{p} \rangle] = \mathbb{E}[(a\boldsymbol{p} + \boldsymbol{t})\, a] = \mathbb{E}[a^2]\, \boldsymbol{p} + \boldsymbol{\gamma}$, and $\boldsymbol{p}^\top M \boldsymbol{p} = \mathbb{E}[a^2]$ because $\boldsymbol{\gamma} \perp \boldsymbol{p}$. Since $\boldsymbol{\gamma}$ is orthogonal to $\boldsymbol{p}$, $M\boldsymbol{p}$ is parallel to $\boldsymbol{p}$ if and only if $\boldsymbol{\gamma} = \boldsymbol{0}$.

(ii) Let $\rho = \mathbb{E}[a^2]$. Since $\lambda_1 \geq \boldsymbol{p}^\top M \boldsymbol{p} = \rho > \lambda_2$, the eigenvalue λ_1 is simple and $\boldsymbol{v}_1$ is unique up to sign. By (i), the residual of $\boldsymbol{p}$ as an approximate eigenvector of M is $\boldsymbol{r} = M\boldsymbol{p} - \rho\boldsymbol{p} = \boldsymbol{\gamma}$. Write $\boldsymbol{p} = \cos\varphi\, \boldsymbol{v}_1 + \sin\varphi\, \boldsymbol{w}$ with $\varphi = \angle(\boldsymbol{v}_1, \boldsymbol{p})$ and a unit vector $\boldsymbol{w}$ in the span of the eigenvectors $\boldsymbol{v}_2, \boldsymbol{v}_3, \ldots$ of M. Because M maps that span into itself,

$$\|\boldsymbol{r}\|^2 = \cos^2\varphi\, (\lambda_1 - \rho)^2 + \sin^2\varphi\, \|(M - \rho I)\boldsymbol{w}\|^2 \;\geq\; \sin^2\varphi\, (\rho - \lambda_2)^2,$$

since every eigenvalue λ_i, $i \geq 2$, satisfies $\lambda_i \leq \lambda_2 < \rho$ and hence $|\lambda_i - \rho| \geq \rho - \lambda_2$. Taking square roots gives $\sin\varphi \leq \|\boldsymbol{\gamma}\|/(\rho - \lambda_2)$, the residual form of the Davis–Kahan $\sin\Theta$ theorem (Davis and Kahan, 1970).

(iii) By Weyl's inequality for the sum $M = \boldsymbol{\mu}\boldsymbol{\mu}^\top + \Sigma$, $\lambda_2(M) \leq \lambda_2(\boldsymbol{\mu}\boldsymbol{\mu}^\top) + \lambda_1(\Sigma) = \|\Sigma\|_{\mathrm{op}}$, because the rank-one matrix $\boldsymbol{\mu}\boldsymbol{\mu}^\top$ has second eigenvalue zero.

Replacing λ_2 by the larger $\|\Sigma\|_{\mathrm{op}}$ in the denominator of (ii) can only increase the bound.

(iv) For every unit vector $\boldsymbol{y}$, the Cauchy–Schwarz inequality gives $\langle \boldsymbol{\gamma}, \boldsymbol{y} \rangle = \mathbb{E}[(a - \bar{R})\langle \boldsymbol{t}, \boldsymbol{y} \rangle] \le \sqrt{\mathrm{Var}(a)}\, \sqrt{\mathrm{Var}\langle \boldsymbol{t}, \boldsymbol{y} \rangle}$. The tangent part is $\boldsymbol{t} = P\boldsymbol{u}$ with the orthogonal projection $P = I - \boldsymbol{p}\boldsymbol{p}^{\top}$, so $\mathrm{Var}\langle \boldsymbol{t}, \boldsymbol{y} \rangle = (P\boldsymbol{y})^{\top} \Sigma\, (P\boldsymbol{y}) \le \|\Sigma\|_{\mathrm{op}}$. Choosing $\boldsymbol{y} = \boldsymbol{\gamma}/\|\boldsymbol{\gamma}\|$ gives $\|\boldsymbol{\gamma}\| \le \sqrt{\mathrm{Var}(a)\, \|\Sigma\|_{\mathrm{op}}}$.

(v) Taking the trace of $M = \boldsymbol{\mu}\boldsymbol{\mu}^{\top} + \Sigma$ and using $\operatorname{tr} M = \mathbb{E}\|\boldsymbol{u}\|^2 = 1$ gives $\operatorname{tr} \Sigma = 1 - \bar{R}^2$. Finally, $\boldsymbol{\mu}\boldsymbol{\mu}^{\top}\boldsymbol{p} = \bar{R}^2\boldsymbol{p}$, so by (i) $\Sigma\boldsymbol{p} = M\boldsymbol{p} - \bar{R}^2\boldsymbol{p} = (\mathbb{E}[a^2] - \bar{R}^2)\, \boldsymbol{p} + \boldsymbol{\gamma} = \mathrm{Var}(a)\, \boldsymbol{p} + \boldsymbol{\gamma}$, using $\mathbb{E}[a] = \bar{R}$; because $\boldsymbol{\gamma} \perp \boldsymbol{p}$, $\boldsymbol{p}^{\top}\Sigma\boldsymbol{p} = \mathrm{Var}(a)$. $\square$

Appendix B. Supplementary tables

Tables B.1 to B.4 give the geometry of every source of the four cohorts together with the change in reconstruction error under the main edits.

Table B.1: Per-source values, PD-UCSD (released tokenizer). $\bar{R}$ with 95% subject-cluster bootstrap interval; colat.: mean colatitude; $\sin\angle(\boldsymbol{p}, \boldsymbol{v}_1)$ with the residual bound (4); $\|P_{16}\boldsymbol{p}\|^2$ as in Table 3; var.: share of the centered variance along the anchor; ΔMSE in percent of the unedited error for zero-anchor (ZA), colatitude +20° (C+), mean-anchor (MA), matched rotation in the support (MR) and tangent resample (TR); K: share of the zero-anchor damage reproduced through the decoder's keys alone, relative to the damage of the full edit.

s	$\bar{R}$	**colat.** (°)	$\sin\angle(\boldsymbol{p}, \boldsymbol{v}_1)$	**bound**	$\|P_{16}\boldsymbol{p}\|^2$	**var.**	**ZA**	**C+**	**MA**	**MR**	**TR**	**K**
1	0.67 [0.66, 0.68]	47.3	0.023	0.025	0.31	2.2%	3.6	1.7	−0.03	1.7	9.2	0.86
2	0.60 [0.59, 0.61]	52.6	0.029	0.032	0.35	2.3%	3.5	1.9	−0.20	1.1	7.8	0.89
3	0.60 [0.59, 0.61]	52.6	0.040	0.043	0.39	2.9%	6.1	3.2	−0.12	1.4	12.5	0.91
4	0.66 [0.65, 0.68]	47.5	0.051	0.053	0.27	3.5%	8.9	4.1	0.18	3.7	23.9	0.96
5	0.73 [0.71, 0.74]	42.8	0.011	0.012	0.47	2.3%	7.0	2.4	0.01	3.5	10.1	0.88
6	0.60 [0.59, 0.62]	52.5	0.037	0.040	0.36	2.6%	5.7	2.5	−0.02	1.4	11.1	0.90
7	0.63 [0.62, 0.65]	49.9	0.035	0.037	0.83	3.5%	4.8	2.2	−0.09	1.3	7.8	0.86
8	0.63 [0.62, 0.65]	50.0	0.024	0.025	0.39	2.6%	6.7	2.7	−0.26	1.8	10.3	0.90
9	0.61 [0.60, 0.63]	51.6	0.023	0.024	0.47	2.6%	5.5	2.4	−0.13	1.3	9.5	0.90
10	0.60 [0.58, 0.61]	52.6	0.036	0.038	0.61	3.1%	4.4	2.1	−0.13	1.2	10.8	0.92
11	0.59 [0.58, 0.61]	52.9	0.023	0.026	0.69	2.9%	4.3	1.7	−0.10	1.1	8.8	0.91
12	0.63 [0.62, 0.64]	50.3	0.030	0.032	0.38	2.7%	4.4	2.1	−0.18	1.6	10.3	0.90
13	0.57 [0.55, 0.58]	55.1	0.025	0.028	0.47	2.3%	5.0	2.5	−0.10	1.1	9.8	0.94
14	0.69 [0.68, 0.70]	45.6	0.019	0.021	0.24	2.3%	5.8	1.9	0.03	2.0	10.3	0.87
15	0.61 [0.60, 0.62]	52.0	0.023	0.026	0.37	2.4%	4.1	1.7	−0.10	1.1	8.6	0.90
16	0.69 [0.68, 0.70]	45.4	0.020	0.020	0.48	2.6%	3.6	1.5	−0.10	2.0	7.4	0.85

Table B.2: Per-source values, PD-Iowa (released tokenizer). $\bar{R}$ with 95% subject-cluster bootstrap interval; colat.: mean colatitude; $\sin\angle(\boldsymbol{p}, \boldsymbol{v}_1)$ with the residual bound (4); $\|P_{16}\boldsymbol{p}\|^2$ as in Table 3; var.: share of the centered variance along the anchor; ΔMSE in percent of the unedited error for zero-anchor (ZA), colatitude +20° (C+), mean-anchor (MA), matched rotation in the support (MR) and tangent resample (TR); K: share of the zero-anchor damage reproduced through the decoder's keys alone, relative to the damage of the full edit.

s	$\bar{R}$	**colat.** (°)	$\sin\angle(\boldsymbol{p}, \boldsymbol{v}_1)$	**bound**	$\|P_{16}\boldsymbol{p}\|^2$	**var.**	**ZA**	**C+**	**MA**	**MR**	**TR**	**K**
1	0.67 [0.66, 0.68]	47.5	0.022	0.023	0.37	2.4%	3.2	1.4	−0.11	1.7	7.6	0.86
2	0.59 [0.59, 0.60]	52.9	0.030	0.033	0.47	2.8%	1.1	0.5	−0.07	0.2	1.7	0.66
3	0.61 [0.60, 0.62]	51.8	0.024	0.027	0.31	2.5%	5.7	2.6	−0.15	1.4	11.7	0.92
4	0.68 [0.67, 0.68]	46.7	0.031	0.033	0.39	2.9%	9.6	3.9	−0.18	4.0	23.0	0.96
5	0.70 [0.69, 0.70]	45.2	0.017	0.018	0.30	2.6%	7.5	3.2	−0.19	3.5	15.5	0.90
6	0.62 [0.60, 0.63]	51.4	0.031	0.034	0.40	3.0%	4.7	2.1	−0.14	1.4	12.1	0.87
7	0.63 [0.62, 0.64]	50.3	0.029	0.032	0.81	3.5%	1.2	0.4	−0.01	0.1	2.3	0.49
8	0.62 [0.61, 0.63]	51.1	0.027	0.029	0.36	2.8%	6.5	3.0	−0.32	1.9	13.5	0.90
9	0.60 [0.59, 0.61]	52.4	0.029	0.032	0.43	3.0%	3.9	1.7	−0.06	1.0	8.4	0.86
10	0.58 [0.57, 0.59]	53.6	0.034	0.038	0.87	3.8%	3.8	1.9	−0.16	0.9	7.7	0.89
11	0.61 [0.60, 0.62]	52.1	0.023	0.025	0.36	2.7%	1.5	0.7	−0.04	0.1	2.7	0.71
12	0.64 [0.63, 0.64]	49.9	0.026	0.028	0.41	3.0%	4.8	2.3	−0.21	2.3	13.1	0.91
13	0.59 [0.58, 0.60]	53.4	0.027	0.029	0.40	2.4%	3.4	1.2	−0.07	1.0	7.0	0.90
14	0.68 [0.66, 0.68]	46.8	0.025	0.028	0.46	2.9%	6.5	2.4	−0.24	2.1	11.9	0.89
15	0.63 [0.62, 0.64]	50.6	0.021	0.023	0.25	2.4%	2.3	0.9	−0.01	0.6	4.5	0.80
16	0.65 [0.64, 0.66]	48.7	0.022	0.024	0.45	3.0%	3.6	1.6	−0.14	1.7	8.2	0.87

Table B.3: Per-source values, Dementia (released tokenizer). $\bar{R}$ with 95% subject-cluster bootstrap interval; colat.: mean colatitude; $\sin\angle(\boldsymbol{p}, \boldsymbol{v}_1)$ with the residual bound (4); $\|P_{16}\boldsymbol{p}\|^2$ as in Table 3; var.: share of the centered variance along the anchor; ΔMSE in percent of the unedited error for zero-anchor (ZA), colatitude +20° (C+), mean-anchor (MA), matched rotation in the support (MR) and tangent resample (TR); K: share of the zero-anchor damage reproduced through the decoder's keys alone, relative to the damage of the full edit.

s	$\bar{R}$	**colat.** (°)	$\sin\angle(\boldsymbol{p}, \boldsymbol{v}_1)$	**bound**	$\|P_{16}\boldsymbol{p}\|^2$	**var.**	**ZA**	**C+**	**MA**	**MR**	**TR**	**K**
1	0.65 [0.64, 0.66]	48.9	0.020	0.021	0.35	2.5%	4.9	2.3	−0.19	1.7	10.0	0.85
2	0.57 [0.56, 0.58]	54.5	0.044	0.047	0.49	3.1%	4.0	2.4	−0.33	0.9	9.0	0.88
3	0.55 [0.54, 0.56]	56.4	0.034	0.039	0.80	3.1%	9.2	5.5	−0.54	1.5	18.8	0.96
4	0.61 [0.60, 0.63]	51.3	0.036	0.041	0.88	3.4%	19.5	10.0	−0.39	4.3	36.3	0.98
5	0.63 [0.62, 0.64]	50.2	0.042	0.046	0.57	3.9%	13.5	6.9	0.10	3.5	24.4	0.94
6	0.51 [0.50, 0.53]	58.6	0.049	0.055	0.53	2.9%	5.9	3.9	−0.43	1.0	13.1	0.93
7	0.58 [0.56, 0.59]	53.8	0.069	0.074	0.78	4.8%	5.4	2.9	−0.57	1.0	10.2	0.89
8	0.54 [0.53, 0.55]	56.8	0.039	0.044	0.30	2.6%	8.3	4.7	−0.42	1.0	12.1	0.93
9	0.56 [0.55, 0.58]	55.0	0.063	0.067	0.54	4.0%	5.2	3.1	−0.07	0.9	11.8	0.91
10	0.53 [0.51, 0.55]	57.2	0.073	0.079	0.85	4.9%	7.0	4.6	−1.01	1.6	17.2	0.95
11	0.49 [0.48, 0.50]	60.2	0.059	0.070	0.39	3.3%	6.2	4.1	−0.14	0.9	13.9	0.96
12	0.56 [0.55, 0.57]	55.3	0.025	0.028	0.57	3.1%	6.3	3.6	−0.37	1.5	13.8	0.91
13	0.52 [0.52, 0.54]	57.9	0.032	0.038	0.52	2.7%	6.6	3.9	−0.24	1.1	12.7	0.95
14	0.62 [0.60, 0.63]	51.1	0.043	0.049	0.26	3.2%	10.4	4.7	−0.31	2.1	19.9	0.94
15	0.54 [0.53, 0.55]	56.8	0.030	0.035	0.87	2.9%	6.4	3.4	−0.11	1.1	11.7	0.95
16	0.61 [0.60, 0.62]	51.3	0.032	0.034	0.85	3.5%	5.0	2.5	−0.22	1.9	12.1	0.89

Table B.4: Per-source values, MDD-MEG (released tokenizer). $\bar{R}$ with 95% subject-cluster bootstrap interval; colat.: mean colatitude; $\sin\angle(\boldsymbol{p}, \boldsymbol{v}_1)$ with the residual bound (4); $\|P_{16}\boldsymbol{p}\|^2$ as in Table 3; var.: share of the centered variance along the anchor; ΔMSE in percent of the unedited error for zero-anchor (ZA), colatitude +20° (C+), mean-anchor (MA), matched rotation in the support (MR) and tangent resample (TR); K: share of the zero-anchor damage reproduced through the decoder's keys alone, relative to the damage of the full edit.

s	$\bar{R}$	**colat.** (°)	$\sin\angle(\boldsymbol{p}, \boldsymbol{v}_1)$	**bound**	$\|P_{16}\boldsymbol{p}\|^2$	**var.**	**ZA**	**C+**	**MA**	**MR**	**TR**	**K**
1	0.71 [0.70, 0.71]	44.7	0.015	0.016	0.40	1.5%	15.7	5.3	0.01	10.4	35.5	0.98
2	0.68 [0.67, 0.69]	46.8	0.017	0.018	0.40	1.7%	10.2	2.5	−0.01	5.6	22.4	0.98
3	0.62 [0.62, 0.63]	51.1	0.020	0.023	0.30	1.4%	7.6	2.7	−0.03	2.4	17.6	0.98
4	0.70 [0.69, 0.70]	45.5	0.012	0.013	0.29	1.3%	9.9	3.2	−0.01	4.5	17.4	0.98
5	0.69 [0.69, 0.70]	45.8	0.021	0.023	0.23	1.4%	9.9	3.0	0.03	4.1	18.9	0.97
6	0.67 [0.66, 0.68]	47.6	0.017	0.019	0.34	1.6%	5.7	1.2	−0.04	2.3	11.2	0.95
7	0.68 [0.67, 0.68]	46.9	0.017	0.019	0.33	1.6%	4.6	1.4	−0.01	2.1	9.9	0.95
8	0.63 [0.62, 0.63]	50.8	0.026	0.029	0.35	1.8%	2.8	1.0	−0.03	0.7	3.7	0.92
9	0.57 [0.57, 0.58]	54.6	0.029	0.034	0.30	1.4%	1.3	0.6	0.03	0.3	2.4	0.83
10	0.66 [0.66, 0.67]	48.2	0.018	0.019	0.29	1.4%	5.0	1.0	0.04	2.6	13.3	0.96
11	0.63 [0.63, 0.64]	50.3	0.017	0.019	0.33	1.4%	11.1	3.3	−0.03	4.1	27.8	0.99
12	0.68 [0.68, 0.69]	46.5	0.015	0.016	0.28	1.4%	14.8	5.3	−0.11	7.7	29.8	0.98
13	0.64 [0.63, 0.65]	50.0	0.016	0.018	0.13	1.5%	5.8	1.4	−0.05	1.8	8.4	0.97
14	0.69 [0.68, 0.69]	46.0	0.024	0.026	0.26	1.6%	8.6	2.4	−0.04	3.4	14.4	0.98
15	0.61 [0.61, 0.62]	51.9	0.023	0.027	0.33	1.4%	2.3	0.7	−0.03	0.6	4.7	0.94
16	0.71 [0.71, 0.72]	44.3	0.012	0.013	0.31	1.3%	9.9	3.3	0.00	7.7	24.2	0.97

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